



DEPARTMENT OF LABOR AND EMPLOYMENT

Division of Workers' Compensation

WORKERS' COMPENSATION RULES OF PROCEDURE WITH TREATMENT GUIDELINES

RULE 17, EXHIBIT 8 CERVICAL SPINE INJURY MEDICAL TREATMENT GUIDELINES

7 CCR 1101-3 Rule 17 - Exhibit 8

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

Section 1. Guidelines Introduction

Section 1.a. Context and Use

This document has been prepared by the Colorado Department of Labor and Employment, Division of Workers' Compensation (Division) and should be interpreted within the context of guidelines for physicians/providers treating individuals who qualify as injured workers with cervical spine injuries under the Colorado Workers' Compensation Act.

Although the primary purposes of this document for practitioners are advisory and educational, these guidelines are enforceable under the Workers' Compensation Rules of Procedure, 7 CCR 1101-3. The Division recognizes that acceptable medical practice may include deviations from these guidelines, as individual cases dictate. Therefore, these guidelines are not relevant as evidence of a provider's legal standard of professional care.

To properly utilize this document, the reader should not skip or overlook any sections.

Section 1.b. Application of the Guidelines

The Division provides procedures to implement medical treatment guidelines (MTGs) and to foster communication to resolve disputes among the provider, payer, and patient through the Workers' Compensation Rules of Procedure. In lieu of more costly litigation, parties may wish to seek administrative dispute resolution services through the Division or the Office of Administrative Courts.

Section 1.c. Guidelines Recommendations and Inclusion of Medical Evidence

All recommendations are based on available evidence and/or consensus judgment. A Division staff methodologist (MD, MSPH) researched and adopted literature critique criteria for certain study designs (e.g., randomized controlled trial [RCT]). Literature critiques were performed in a manner congruent with national standards and were completed independent of the multidisciplinary task force group that drafted initial recommendations. The methodology is described in detail on the Division's MTGs page at cdle.colorado.gov/medical-treatment-guidelines. Please also refer to the Division's website for evidence tables and study critiques that provide details on the studies used to develop the evidence statements.

When possible, guidelines recommendations link to evidence tables that include the level of evidence supporting the treatment recommendation. An evidence summary is included in tables where multiple evidence statements address the same topic. Some of the evidence tables include literature on low back pain when there is a presumed physiologic similarity for neck pain. It is generally recognized that early reports of a positive treatment effect are frequently weakened or overturned by subsequent research. When interpreting medical evidence statements in the guidelines, the following apply:

- “Some evidence” means the recommendation considered at least 1 adequate scientific study, which reported that a treatment was effective. The Division recognizes that further research is likely to have an impact on the intervention’s effect.
- “Good evidence” means the recommendation considered the availability of multiple adequate scientific studies or at least 1 relevant high-quality scientific study, which reported that a treatment was effective. The Division recognizes that further research may have an impact on the intervention’s effect.
- “Strong evidence” means the recommendation considered the availability of multiple relevant and high-quality scientific studies, which arrived at similar conclusions about the effectiveness of a treatment. The Division recognizes that further research is unlikely to have an important impact on the intervention’s effect.

Recommendations not linked to evidence tables are based on consensus; consensus means the judgment of experienced professionals based on general medical principles.

All recommendations in the guidelines are considered to represent reasonable care in appropriately selected cases, irrespective of the level of evidence or consensus statement attached to them. Those procedures considered inappropriate, unreasonable, or unnecessary are designated in the guidelines as “not recommended.”

Section 1.d. Recommended Citation for This Document

Colorado Division of Workers' Compensation. (2021). Cervical Spine Injury Medical Treatment Guidelines. Colorado Department of Labor and Employment. <https://cdle.colorado.gov/medical-providers/medical-treatment-guidelines>.

Section 1.e. Glossary of Abbreviations

- ACDF - Anterior cervical discectomy and fusion
- ADLs - Activities of daily living
- BBHI 2 - Brief Battery for Health Improvement, 2nd Edition
- BCIA - Biofeedback Certification International Alliance
- BDI-II - Beck Depression Inventory-II
- BHI-MV - Behavioral Health Index-Multimedia Version
- BMP - Bone morphogenetic protein
- CBT - Cognitive behavioral therapy
- CDA - Cervical disc arthroplasty
- CES-D - Center for Epidemiological Studies Depression Scale
- COX-2 - Cyclooxygenase-2
- CT - Computed tomography
- DRAM - Distress and Risk Assessment Method
- DSM - Diagnostic and Statistical Manual of Mental Disorders
- ESI - Epidural steroid injection
- FCE - Functional capacity evaluation
- G6PD - Glucose-6-phosphate-dehydrogenase
- GAD-7 - Generalized Anxiety Disorder Scale
- GI - Gastrointestinal
- H2-blocker - Histamine H2-receptor antagonists
- MME - Morphine milligram equivalents
- MMI - Maximum medical improvement
- MRI - Magnetic resonance imaging
- MTGs - Medical treatment guidelines
- NMES - Neuromuscular electrical stimulation
- NRS - Numeric rating scale
- NSAIDs - Non-steroidal anti-inflammatory drugs
- ODI - Oswestry Disability Index, or Oswestry Low Back Pain Disability Questionnaire

- ORT - Opioid Risk Tool
- PDMP - Prescription Drug Monitoring Program
- PEEK - Polyetheretherketone
- PENS - Percutaneous electrical nerve stimulation
- PHQ - Patient Health Questionnaire
- PRIME-MD - Primary Care Evaluation for Mental Disorders
- QPOP - Quality Performance and Outcomes Payments
- RCT - Randomized controlled trial
- RF - Radiofrequency
- rhBMP - Recombinant human bone morphogenetic protein
- rhBMP-2 - Recombinant human bone morphogenetic protein-2
- ROM - Range of motion
- SF-12 - 12-item Short Form Health Survey
- SF-36 - 36-item Short Form Health Survey
- SI - Sacroiliac
- SNRI - Serotonin norepinephrine reuptake inhibitor
- SSRI - Selective serotonin reuptake inhibitors
- TDR - Total disc replacement
- TENS - Transcutaneous electrical nerve stimulation
- VAS - Visual analogue scale

Section 2. General Guidelines Principles

This document should be interpreted within the parameters of the following guidelines principles that may lead to more optimal medical and functional outcomes for injured workers.

Section 2.a. Education

Education of the individual and family and/or support system, as well as the employer, insurer, policy makers, and the community, should be the primary emphasis in the treatment of cervical spine injury and disability. Currently, practitioners often think of education last, after medications, manual therapy, and surgery. Practitioners must implement strategies to educate individuals with cervical spine injury, employers, insurance systems, policy makers, and the community as a whole. An education-based paradigm should always start with inexpensive communication that provides recovery, function-focused, patient-centered, and evidence-based information to the individual with cervical spine injury. More in-depth education is currently a component of treatment regimens that employ functional, restorative, preventive, and rehabilitative programs. No treatment plan is complete without addressing issues of individual and/or group patient education as a means of facilitating self-management of symptoms and prevention. Facilitation through language interpretation, when necessary, is a priority and part of the medical care treatment protocol.

Section 2.b. Shared Decision Making

Providers should implement shared decision making as a crucial element of a successful treatment plan. Patients, with the assistance of their health care practitioner and support system, should identify their personal and professional functional goals of treatment at the first visit. Progress towards the individual's identified functional goals should be addressed by all members of the health care team at subsequent visits and throughout the established treatment plan. Nurse case managers, psychologists, physical therapists, and other members of the health care team play an integral role in shared decision making and achievement of functional goals. Patient education and shared decision making should facilitate self-management of symptoms and prevention of further injury.

Section 2.c. Return to Work

Return to work is therapeutic, assuming the work is not likely to aggravate the basic problem. The practitioner must provide specific written physical limitations, and the patient should never be released to work with non-specific and vague descriptions such as "sedentary" or "light duty." The following physical limitations should be considered and modified as recommended: lifting, pushing, pulling, crouching, carrying, walking, using stairs, bending at the waist, awkward and/or sustained postures, tolerance for sitting or standing, hot and cold environments, repetitive motion tasks, sustained grip, tool usage, and vibration factors. Even if there is residual chronic pain, return to work is not usually contraindicated. The practitioner should understand all of the physical demands of the patient's job position before returning the patient to full duty and should request clarification of the patient's job duties. Clarification should be obtained from the employer or, if necessary, including, but not limited to, an occupational health nurse, occupational therapist, vocational rehabilitation specialist, or an industrial hygienist.

Section 2.d. Treatment Parameter Duration

Time frames for specific interventions commence once treatments have been initiated, not on the date of injury. Duration will be impacted by the individual's adherence, as well as availability of services. Clinical judgment may substantiate the need to accelerate or decelerate the time frames discussed in this document.

Section 2.e. Active Interventions

Active interventions emphasizing patient responsibility, such as therapeutic exercise and/or functional treatment, are generally utilized over passive interventions, especially as treatment progresses. Generally, passive interventions are viewed as a means to facilitate progress in an active rehabilitation program with concomitant attainment of objective functional gains.

Section 2.f. Active Therapeutic Exercise Program

Exercise program goals should incorporate patient strength, endurance, flexibility, coordination, and education. This includes functional application in vocational or community settings.

Section 2.g. Positive Patient Response

Positive results are defined primarily as functional gains that can be objectively measured. Objective functional gains include, but are not limited to: positional tolerances, range of motion (ROM), strength, endurance, activities of daily living (ADLs), ability to function at work, cognition and communication, psychological behavior, and efficiency/velocity measures that can be quantified. Subjective reports of pain and function should be considered and given relative weight when the pain has anatomic and physiologic correlation. Anatomic correlation must be based on objective findings. Patient completed functional questionnaires can provide useful additional confirmation, such as those recommended by the Colorado Division of Workers' Compensation (Division) Quality Performance and Outcomes Payments program (QPOP, see Rule 18), the Patient-Specific Functional Scale, or other validated function scales.

Section 2.h. Re-evaluation of Treatment Effectiveness

Re-evaluation should occur every 3 to 4 weeks or within the time to produce effect for a given treatment. Treatment should be modified or discontinued if there is no evidence of positive results. Before discontinuing the treatment, the provider should have a detailed discussion with the patient to determine the reason for failure to produce positive results. Reconsideration of diagnosis should also occur in the event of a poor response to a seemingly rational intervention.

Section 2.i. Surgical Interventions

Surgery should be contemplated within the context of expected functional outcome and not purely for the purpose of pain relief. The concept of "cure" with respect to surgical treatment by itself is generally a misnomer. All operative interventions must be based upon positive correlation of clinical findings, clinical course, and diagnostic tests. A comprehensive assimilation of these factors must lead to a specific diagnosis with positive identification of pathologic conditions.

Section 2.j. 6-month Time Frame

The prognosis drops precipitously for returning an injured worker to work once they have been temporarily totally disabled for more than 6 months. The emphasis within these guidelines is to move patients along a continuum of care and return to work within a 6-month time frame, whenever possible. It is important to note that time frames may be less pertinent for injuries that do not involve work-time loss or are not occupationally related.

Section 2.k. Delayed Recovery

For patients who are failing to make expected progress 6 to 12 weeks after initiation of treatment of an injury, strongly consider a psychological evaluation, if not previously provided, as well as initiating interdisciplinary rehabilitation treatment and vocational goal setting. The Division recognizes that 3 to 10% of all industrially injured patients will not recover within the timelines outlined in this document, despite optimal care. Such individuals may require treatments beyond the timelines discussed within this document, but such treatment requires clear documentation by the authorized treating practitioner focusing on objective functional gains afforded by further treatment and impact upon prognosis.

Section 2.l. Post Maximum Medical Improvement (MMI) Care

Maximum medical improvement (MMI) should be declared when a patient's condition has plateaued to the point where the authorized treating physician no longer believes further medical intervention is likely to result in improved function. However, some patients may require treatment after MMI has been declared in order to maintain their functional state. The recommendations in these guidelines are for pre-MMI care and are not intended to limit post-MMI treatment.

Section 3. Overview of Care

Introduction. Neck pain is a common condition in the adult population. Neck pain in the working population most commonly occurs from cervical strain injuries, whiplash associated disorder, or degenerative conditions aggravated by work. Most individuals with neck pain without neurological findings will recover with therapy and self-management without invasive measures. These guidelines outline a biopsychosocial approach to neck pain care and integrate biological, psychological, and social elements.

Significant trauma resulting in fractures and/or spinal cord dysfunction are not covered in these guidelines. Early imaging and surgical evaluation should be reserved for more severe issues outlined in [recommendation 1](#). Most patients will respond to conservative care as outlined in recommendations 2 through 5 and will not require invasive treatment. Additional interventions, as indicated for less common cases, are covered by the remaining recommendations. Refer to the [Mild Traumatic Brain Injury Medical Treatment Guidelines \(MTGs\)](#) for information on cervicogenic headache.

Recommendations.

Core Requirements.

Recommendation 1. Early imaging and surgical evaluation is required for patients with signs of myelopathy, progressive neurologic deficits, upper extremity weakness, or epidural abscess.

Recommendation 2. Initial conservative management without imaging is strongly recommended for neck pain patients who do not meet the criteria in [recommendation 1](#), focusing on the following:

- education that prolonged periods of immobility are detrimental,
- education on the positive impact of movement on recovery,
- appropriate use of over the counter medications, and
- application of ice or heat.

Recommendation 3. Cervical immobilization as the sole treatment is not recommended for nonspecific neck pain when there is no clinical concern for instability.

Recommendation 4. Close follow-up and subsequent neurologic examinations are required for patients with radicular neck pain who do not meet the criteria in [recommendation 1](#) or [recommendation 8](#).

Recommendation 5. Participation in self-directed exercise or an active therapy program that includes core stabilization, strengthening, and endurance is recommended as a principal neck pain treatment. See the Active Therapies section.

Recommendation 6. Individuals with barriers to functional recovery may benefit from an interdisciplinary approach to care. [See the Diagnosis section.](#)

Recommendation 7. Referral for epidural steroid injection (ESI) evaluation within 6 weeks of pain onset is permitted when all of the following criteria are met:

- function-limiting pain in arms greater than the neck that interferes with return to work, activities of daily living (ADLs), and/or participation in active therapy; and
- positive correlation among clinical findings, the clinical course, and diagnostic tests.

See the [Epidural Steroid Injection section](#).

Recommendation 8. Referral for discectomy evaluation within 6 weeks of pain onset is permitted when all of the following criteria are met:

- function-limiting pain in arms greater than the neck that interferes with return to work, ADLs, and/or participation in active therapy;
- physical exam findings of abnormal reflexes, motor weakness, or radicular sensory deficits;
- findings on magnetic resonance imaging (MRI) indicate impingement of nerves or the spinal cord corresponding to reproducible physical exam findings; and
- the observed pathology is amenable to surgical intervention.

See the Discectomy section.

Recommendation 9. Referral for spinal injection is indicated when all of the following criteria are met:

- positive correlation among clinical findings, the clinical course, and diagnostic tests;
- positive functional response to a diagnostic injection, if required;
- persistent functional impairment despite engagement in 6 weeks of active therapy; and

confounding psychosocial risk factors have been screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

See the Injections section.

Recommendation 10. Referral of patients with cervical radiculopathy for surgical evaluation is recommended when confirmatory imaging studies specifically correlate with clinical findings, demonstrating nerve root compression or spinal cord compromise, and any of the following are present:

- acute, incapacitating pain with progressive neurologic deficits;
- persistent or recurrent arm pain with functional limitations unresponsive to 6 weeks of treatment;
- progressive neurologic deficits; or
- static neurologic deficit with significant radicular pain.

Evidence Tables. See related sections for evidence tables.

Section 4. Diagnosis

Introduction. Initial and follow-up diagnostic evaluation of neck pain may include a clinical history, physical examination, laboratory studies, imaging studies, and electrodiagnostic studies. Please see specific sections for diagnostic injections.

Recommendations.

Initial Diagnostic Studies Requirements.

Recommendation 11. A detailed history of pain onset, past medical history, and a detailed neurologic examination are required at the initial neck pain evaluation (tables 1, 2).

Recommendation 12. Medical causation must establish that neck pain is a result of a specific injury, an aggravation of an underlying condition, or a previously asymptomatic condition made symptomatic by a work-related exposure (table 3).

Recommendation 13. Initial functional assessment is strongly recommended. See the Appendix Functional Screens.

Recommendation 14. A psychological screen is encouraged as a routine part of clinical care and is required as soon as any of the following barriers to functional recovery are identified (Appendix Psychological Screens):

- limited patient engagement in recovery,
- activity avoidance or catastrophization due to pain,
- avoidance of essential recovery activities,
- inappropriate expression of a sense of injustice,
- exaggeration of symptoms/situation,
- low expectations of recovery,
- ineffective coping skills, or
- loss of vocational connection.

Individuals with barriers to functional recovery may benefit from an interdisciplinary approach to care (table 4).

Recommendation 15. Imaging studies are recommended if the patient is 55 years or older, if there is a history of significant trauma, if there is persistent or unexplained pain after 6 weeks, or if there is concern for any of the following:

- fracture,
- instability,
- occult cancer,
- weakness,
- pain worse at rest,
- epidural abscess, or
- myelopathy.

Documentation shall include the specific findings under investigation and how the imaging test will influence treatment.

Recommendation 16. Dynamic fluoroscopy is permitted in the initial or follow-up evaluation of cervical spine trauma and in select cases for comprehensive evaluation of neck and radicular pain.

Recommendation 17. When clinically indicated, early laboratory studies may be ordered to evaluate for systemic illness, infection, neoplasia, underlying rheumatologic, or connective tissue disorder.

Follow-Up Diagnostic Studies Requirements.

Recommendation 18. A formal psychological or psychosocial evaluation is required for patients not making expected progress within 6 weeks of injury and whose subjective symptoms do not correlate with objective signs and tests. See the Behavioral and Psychological Interventions section.

Recommendation 19. Magnetic resonance imaging (MRI) is indicated when there is concern for myelopathy, masses, infection, metastatic disease, or cord contusion. MRI is recommended for patients who have responded poorly to initial care and there is clinical suspicion for disc herniation or nerve root compression. Contrast MRI can be used when there is a history of cervical surgery or if there is possible infection, malignancy, or tumor (table 5).

Recommendation 20. Computed tomography (CT) scan is permitted to better visualize bone and to further evaluate masses and suspected fractures not clearly identified on radiographic evaluation.

Recommendation 21. Myelography is a diagnostic option when:

- CT or MRI are contraindicated or not available, and
- when other tests have proven non-diagnostic in the surgical candidate.

Recommendation 22. Radioisotope bone scanning is permitted when there is clinical suspicion for metastatic or primary bone tumors, occult or stress fractures, osteomyelitis, infection, or other inflammatory lesions.

Recommendation 23. Discography is not recommended for use in evaluating cervical spine conditions. See the Spinal Fusion section.

Recommendation 24. Electromyography and nerve conduction studies are recommended for patients with suspected neural involvement and persistent symptoms that are unresponsive to initial conservative treatment.

Recommendation 25. Somatosensory evoked potentials may be used to evaluate myelopathy, neurogenic bladder, and sexual dysfunction, but it is not recommended to identify radiculopathy.

Recommendation 26. Surface electromyography and current perception threshold evaluation are not recommended.

Evidence Tables.

Table 1.		
<i>Evidence Table: Self-reported Body Pain Diagrams in Describing Neck Pain</i>		
Summary: Pain diagrams, either manual or computerized, reliably identify pain location.		
Some evidence	Evidence statement	Design
	Using a computer program (electronic body surface estimation method) to measure pain distribution was found to have both high intra- and inter-examiner reliability in patients with chronic nonmalignant pain.	Systematic review
	A body diagram, in which the patients use a pencil to shade in painful areas of the body, has acceptable reliability for designating the regions in which their pain is felt.	Systematic review

Table 2.		
<i>Evidence Table: Physical Examination Findings and Neck Pain</i>		
Summary: The findings on physical examination may be clinically suggestive of neurologic compression.		
Some evidence	Evidence statement	Design
	Spurling's test, traction/neck distraction, and Valsalva demonstrate reasonable specificity. With high suspicion for nerve root compression, they are helpful to clinically rule in radiculopathy.	Systematic review
Table 2 continued.		
<i>Evidence Table: Physical Examination Findings and Neck Pain</i>		
Some evidence cont.	The upper limb tension test should be done with finger and wrist extension. When performed in this manner and with a low index of clinical suspicion, a negative upper limb tension test may be helpful to rule out radiculopathy.	Systematic review

Table 3.		
<i>Evidence Table: Occupational Risk Factors for Neck Pain</i>		
Summary: Neck pain is caused by a combination of workplace factors (e.g., poor workstation ergonomic design, work posture, and workplace activities) and personal factors (e.g., age, previous musculoskeletal pain, and low physical fitness). Prolonged neck flexion while performing repetitive or precision work and performing tasks that require sustained trapezium activity may result in neck pain.		
Strong evidence	Evidence statement	Design
	Neck pain in the workplace is multifactorial, and a combination of workplace and individual factors is necessary to cause neck pain.	Systematic review

Some evidence	Evidence statement	Design
	Repetitive or precision work, accompanied by prolonged neck flexion, are likely risk factors for neck pain in the workplace.	Systematic review
	Sustained trapezius muscle activity predicts later neck and shoulder pain.	Cohort study

Table 4.

Evidence Table: Interdisciplinary Neck Pain Care

Good evidence	Evidence statement	Design
	Interdisciplinary programs that include screening for psychological issues, identification of fear-avoidance beliefs and treatment barriers, and establishment of individual functional and work goals will improve function and decrease disability.	RCT

Table 5.

Evidence Table: Imaging Findings and Neck Pain

Summary:

Imaging findings occur frequently in asymptomatic patients and therefore are a poor indicator for clinically significant pathology.

Table 5 continued.

Evidence Table: Imaging Findings and Neck Pain

Some evidence	Evidence statement	Design
	The majority of workers over 60 show evidence of disc degeneration on cervical spine MRI, and posterior disc protrusions are present in the majority of asymptomatic workers over 40 years of age.	Cohort study
	Asymptomatic subjects of all ages frequently show common degenerative changes in the cervical and lumbar spine as seen on MRIs. These changes are more common with increasing age. The cervical and lumbar spine are equally affected, suggesting that disc degeneration occurs in tandem in the lumbar and cervical spine.	Cohort study

Section 5. Return to Activity and Work Considerations

Introduction. Return to activity and/or work-related activities is a primary therapeutic component of neck pain management and rehabilitation. There are special tests performed as a part of a skilled assessment of the patient's capacity to return to work, their strength capacities, and/or physical work demand classifications and tolerance. Terminology relevant to the activity and return to work discussion includes the following:

- computer enhanced evaluations,
- immobility,
- functional capacity evaluation (FCE),
- jobsite evaluation or analysis, and
- work tolerance screenings.

Communication among the patient, treating provider, employer, and insurer is essential for successful return to work efforts. Refer to the [Active Therapies section](#) for work conditioning and work simulation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Return to Work.

- Absolute and relative contraindications include performance of job activities that pose a medical risk to the recovering worker.

Side Effects and Adverse Events Related to Return to Work.

- Side effects and adverse events include a temporary increase in discomfort or muscle soreness when correcting deconditioning after time away from typical tasks.

Recommendations.

Core Requirements.

Recommendation 27. A job history interview should be completed at the time of the initial evaluation. A thorough job history generally includes:

- duties and demands of job held at time of injury;
- duties and demands of current job, if different;
- stressors; and
- cognitive and social issues, with treatment incorporated into the plan of care.

Recommendation 28. A formal job description for the injured worker is required to identify physical demands at work and to assist in the creation of modified duty.

Recommendation 29. Ergonomic or adaptive equipment, therapeutic breaks, and workplace interventions are recommended to maintain employment ([table 6](#)).

Recommendation 30. Nurse case management is recommended in medically complex cases to facilitate communication between the primary provider, referral providers, insurer, employer, and employee. The patient may decline nurse case management services.

Workplace Tests Requirements.

Recommendation 31. A jobsite evaluation is permitted to fully understand the physical demands of a working day; this information can be used when determining medical causation, the need for ergonomic changes, and/or ability to return to work.

Recommendation 32. A work tolerance screening is recommended to determine a patient's tolerance for performing a specific job activity or task from a cardiovascular, postural tolerance, and physical fitness perspective.

Recommendation 33. An FCE may be used to determine an individual's capacity to work. A formal job description and jobsite evaluation, if performed, should be made available to the FCE evaluator prior to having the FCE performed.

Recommendation 34. Follow-up evaluation with the treating therapist and/or the treating physician is required between 1 to 3 days after the FCE to assess the patient's status.

Recommendation 35. FCEs are not recommended as the sole tool for the development of temporary or permanent work restrictions (table 7).

Recommendation 36. Performance during FCEs cannot be used as the sole criteria in diagnosing malingering.

Recommendation 37. Computer-enhanced strength evaluations or performance on FCEs cannot be used alone to determine return to work restrictions.

Return to Work Requirements.

Recommendation 38. It is strongly recommended that the patient return to work in a modified capacity as soon as it is medically appropriate.

Recommendation 39. Interdisciplinary services are permitted to assist the injured worker in return to work efforts (e.g., behavioral and/or psychological support, active therapy).

Recommendation 40. A graduated return to work strategy should be incorporated into a successful medical treatment plan and rehabilitation after injury with a goal of return to full duty, if medically feasible (e.g., nature and frequency of activities, and hours worked).

Recommendation 41. Permanent work restrictions should be developed based on objective information available, including:

- history;
- findings on physical examination and diagnostic testing; and
- functional response to active therapy, work conditioning, and/or modified duty.

Timing of Tests.

Timing of Tests and Maximum Allowed	
FCE	1 time to determine baseline status, and 1 time to determine permanent work restrictions at case closure if it is clear that the patient cannot return to the position held at the time of injury.
Jobsite evaluations	1 time for initial evaluation, 1 for mid-treatment assessment, and 1 at final evaluation.
Computer enhanced evaluations	1 time for initial evaluation, 1 for mid-treatment assessment, and 1 at final evaluation.
Work tolerance screening	1 time for the initial screen. May monitor improvements in strength every 3 to 4 weeks up to a total of 6 visits.

Evidence Tables.

Table 6.		
<i>Evidence Table: Return to Work and Spine Pain</i>		
Summary: For workers with musculoskeletal conditions, including neck pain, workplace interventions such as ergonomic adjustments and work modification reduce the recurrence of further injury and reduce time away from work. These interventions involve the worker, employer, and an occupational physician.		
Strong evidence	Evidence statement	Design
	For workers who have been absent from work due to musculoskeletal conditions, the cumulative time away from work in the first 12 months is likely to be reduced if there is a workplace intervention in which several stakeholders, including the workers, the employer, and an occupational physician, formulate a plan to reduce the recurrence of further injury; this generally will involve an ergonomic evaluation of the worksite and plans for work modification as needed.	Systematic review
Good evidence	Evidence statement	Design
	Workplace interventions in which several stakeholders, including the workers, the employer, and an occupational physician formulate a plan to reduce the recurrence of further injury are also likely to shorten the time needed for a first return to work, and are likely to be beneficial in reducing the risk of a recurrence of sickness absence due to the underlying musculoskeletal condition.	Systematic review
Some evidence	Evidence statement	Design
	An integrated care program, consisting of workplace interventions and graded activity teaching that pain need not limit activity, is effective in returning patients with chronic low back pain to work, even with minimal reported reduction of pain.	RCT

Table 7.

Evidence Table: FCE

Summary:

FCEs are poor predictors of return to work, and there is no evidence that they predict re-injury. A short form FCE, designed for completion in 1 day, may have similar predictive value to standard FCEs.

Good evidence	Evidence statement	Design
	Performance-based measures, such as those assessed in an FCE, are weakly predictive of return to work variables. Most of the variation in time for return to work is explained by factors other than the physical ability to perform specific tasks.	Systematic review
Some evidence	Evidence statement	Design
	Time off work and gender are important predictors for return to work, and floor-to-waist lifting may also help predict return to work, however, the strength of that relationship has not been determined.	Cohort study
	A short form FCE reduced to a few tests produces a similar predictive quality compared to the longer 2-day version of the FCE regarding length of disability and recurrence of a claim after return to work.	RCT

Table 7 continued.

Evidence Table: FCE

Lack of evidence	Lack of evidence statement	Design
	There is a lack of evidence supporting the validity of FCE for prediction of re-injury following return to work.	Systematic review

Section 6. Essential First Line Treatment

Introduction. Education about neck pain and associated disability may involve the patient, patient's family, employer, insurer, policymakers, and community. Informed consent and shared decision making are 2 important components of patient education. Informed consent is an individual's autonomous authorization of a medical intervention that has resulted from a conversation between a patient and provider. It is also a formal process that healthcare institutions require before permitting procedures. Shared decision making involves an exchange of ideas between the patient and provider and collaboration in the decision itself.

Recommendations.

Core Requirements.

Recommendation 42. Patient education is required as a primary component of neck pain treatment, beginning with the expected natural history of neck pain, with and without radicular symptoms ([table 8](#)).

Recommendation 43. In the setting of benign clinical examination, it is strongly recommended that the provider educates and reassures the patient that there is a high likelihood their condition will improve and that it is essential to self-manage their symptoms.

Recommendation 44. Patient education should include a thorough discussion of how behavioral health evaluation and treatment is an essential component to support recovery from physical injury.

Recommendation 45. Shared decision making is required when 2 or more medically reasonable treatment choices exist, regardless of whether the degree of risk is high or low. Elements of shared decision making for spine interventions must include the following:

- the patient's experience with treatment;
- creation of individualized functional goals of treatment and anticipated barriers to success;
- documentation of expected results of diagnostic testing and possible plan of action in response to test results;
- a discussion of the continuum of treatment from the least invasive to the most invasive, with the intent of identifying a treatment along this continuum that most completely addresses the condition;
- expectation regarding the functional impact of the proposed treatment, including a discussion regarding return to work and expected timeframe for treatment;
- criteria for determining treatment success or failure;
- confirmation of the patient's commitment to perform active therapy to optimize treatment outcomes; and
- consideration of the patient's unique risks and benefits based on comorbid medical conditions

Shared decision making discussions should be tailored to the patient's health literacy.

Recommendation 46. Informed consent is required when a high risk treatment is under consideration. It includes the following elements:

- discussion of the purpose of the proposed treatment;
- benefits and risks of the proposed treatment, alternative treatments, and nontreatment; and
- explicit patient agreement or refusal.

Evidence Tables.

Table 8.		
<i>Evidence Table: Education and Shared Decision Making</i>		
	Evidence statement	Design
Some evidence	Structured educational interventions, whether in the form of written material, oral instruction, or videos, are not expected to lead to significant clinical benefits for whiplash or other neck-associated disorders unless it is combined with exercise or individual advice from a physical therapist.	Systematic review

Section 7. Second Line Treatment

Section 7.a. Core Second Line Treatment

Section 7.a.i. Active Therapies

Introduction. Active therapies are therapeutic exercises and/or activities that intend to restore flexibility, strength, endurance, function, and range of motion (ROM) and to alleviate discomfort. Given the anatomic proximity, active therapies for cervical conditions may also incorporate the shoulder. They require intrinsic motivation by the patient to complete a specific exercise or task. The following active therapies are common treatments:

- activities of daily living (ADLs) therapy,
- functional activities therapy,
- functional electrical stimulation or neuromuscular electrical stimulation (NMES),
- neuromuscular re-education,
- pilates,
- tai chi,
- therapeutic exercise,
- yoga,
- work conditioning, and
- work simulation.

Active therapies also include developing functional goal-setting, maintaining or returning to usual activities and exercise in a graded fashion, and providing patient education and key recovery messages. Active therapies may coincide with pain neuroscience education. Pain neuroscience education involves educating patients about the biological and physiological processes involved in their pain experience and, importantly, deemphasizing the issues associated with anatomical structures, pain generators, and tissue damage.

Individual patients may benefit from additional education approaches. Education may include, but is not limited to: a favorable prognosis for recovery, the importance of continuing daily activities, promotion of self-efficacy, problem-solving, engagement of support systems, and relaxation techniques.

Contraindications / Side Effects and Adverse Events.

Relative Contraindications to Active Therapies.

- Motor, sensory, or reflex abnormalities are relative contraindications to the use of the McKenzie Method of Mechanical Diagnosis and Therapy.
- Pulmonary or cardiac conditions limiting participation are relative contraindications to active therapy.

Side Effects and Adverse Events Related to Active Therapies.

- Exercise therapy may result in transient neck, shoulder, or thoracic muscle soreness. This side effect, within reasonable limits, should not deter continued active therapy.

Recommendations.

Core Requirements.

Recommendation 47. Active interventions, including therapeutic exercise and/or functional treatment, are recommended in the treatment of neck pain ([table 9](#)).

Recommendation 48. Medical clearance is required prior to participation in active therapies if a patient has any of the following:

- angina/dyspnea on exertion or at rest,
- paroxysmal nocturnal dyspnea and/or orthopnea,
- syncope or presyncope,
- arrhythmia or palpitations, or
- cardiac murmur (see Contraindications to Active Therapies).

Recommendation 49. Patients in active therapy must:

- demonstrate functional progress that is documented through validated sequential functional assessment measures,
- return to work with decreased restrictions, and/or
- have improvement in clinical measures (e.g., strength, ROM, and ADLs)

If there is no documented evidence of functional progress after 6 treatments, the therapy will be discontinued and the patient must be referred back to their treating provider for further evaluation. Each patient is limited to a maximum of 4 discrete active therapy trials without documented functional progress ([Appendix Functional Screens](#)).

Recommendation 50. A patient is allowed up to 6 active therapy visits to advance their active home exercise program. These visits are contingent on documented demonstration of previously instructed exercises, performance of their home program at the recommended frequency, and progress in their exercise program.

Recommendation 51. Adjunct passive therapy can occur concurrently with active therapy and the frequency of passive therapies will decrease over time. [See the Passive Therapies section.](#)

Recommendation 52. Durations of care beyond those listed as “time to produce effect” and “maximum” are warranted in certain circumstances when treatment to date has resulted in measurable and clinically meaningful functional improvement. These circumstances include:

- re-injury;
- interrupted continuity of care;
- after surgery, particularly after multiple surgeries;
- injuries (e.g., fracture); or
- comorbidities.

Recommendation 53. Functional electrical stimulation or NMES home units require prior authorization, documenting medical justification for home use.

Time Frames.

Time Frames for Active Therapies			
Time to produce effect	Frequency	Optimum duration	Maximum duration
6 treatments	up to 4 times per week	6 weeks	8 weeks

Evidence Tables.

Table 9.

Evidence Summary: Active Therapies

Active therapies may provide functional benefit and symptomatic relief in patients with neck pain. No evidence demonstrates that 1 form of active therapy is superior to any other, and the selection of an active therapy may be guided by local availability and the patient's preference and capability. Patients with barriers to functional recovery may benefit from the incorporation of pain education into therapeutic exercise.

Table 10.

Evidence Table: Exercise and Neck Pain

Summary:

For patients with subacute and chronic neck pain, exercise provides symptomatic relief as compared to control interventions. For patients with neck pain, participation in active therapy in combination with passive therapy (e.g., manipulation, mobilization, stretching, muscle energy) results in reduced pain and disability as compared to either treatment alone, education alone, or other minimal intervention.

Good evidence	Evidence statement	Design
	Adding exercise in combination with other interventions such as: 1) manipulation alone, or 2) manipulation and mobilization, or 3) mobilization, muscle energy, and stretching, is more effective than manipulation alone, mobilization alone, exercise alone, and other minimal intervention or education alone in reducing neck pain and disability.	Systematic review
	In people with chronic neck pain, exercise with or without manipulation results in decreased pain and disability compared to home exercise and advice at 12 weeks but not 52 weeks. The percentage of patients with 2.5% decrease in pain at 1 year was 51% with manipulation and exercise, 57% with exercise only, and 41% with home exercise and advice. It is not clear that the expense is worth the outcome long-term.	RCT
	A 12-week physiotherapist-led neck-specific exercise program (with or without a behavioral approach) consisting of 2 physiotherapy sessions weekly in addition to home-exercise of increasing intensity in terms of duration (or repetitions) was found to be more effective in reducing pain and disability than "prescription of physical activity." The effect was sustained up to 6 months in both groups as well.	RCT

	For patients with subacute to chronic mechanical neck disorders, a combination of scapulothoracic and upper extremity strength training decreased pain intensity compared to control interventions. The programs ranged from 2 minutes to 2 hours per session, sessions ranged from 3 times/week to 6 times/week, and the programs lasted up to 20 weeks. All programs included specific forms of scapulothoracic and upper extremity strength training.	Systematic review
	For patients with chronic mechanical neck disorders a program consisting of repeated active cervical rotations, strength and flexibility exercises had a beneficial effect on reducing pain intensity. The specific programs ranged from 3 to 6 months (the shorter-length program included a 3 month-self-directed follow-up), 18 to 24 sessions, and duration of sessions was 45 to 90 minutes. Exercise was monitored by a qualified physiotherapist.	Systematic review
Table 10 continued.		
<i>Evidence Table: Exercise and Neck Pain</i>		
Good evidence cont.	For patients with chronic mechanical neck disorders, a combination of cervico-scapulothoracic strengthening and stretching exercises improved function at the intermediate term follow-up (defined as 3 months up to, but not including, 1 year). The specific programs ranged from 6 weeks to 11 weeks, the sessions ranged from 1 to 2 times/week, and the duration of sessions ranged from 20 minutes to 45 minutes.	Systematic review
	In the setting of symptomatic cervical radiculopathy from herniated discs and/or osteophytes at 1 or 2 levels, it is reasonable to initiate a program of structured physical therapy before any surgical decisions are made, since many patients will experience significant improvement in pain and function. However, a program of anterior cervical discectomy and fusion (ACDF) plus structured physical therapy is likely to be superior to physical therapy alone, and the improvements appear to continue for at least 5 to 8 years.	RCT
	In a population of female office workers with "constant chronic neck pain and disability" and symptoms lasting greater than 6 months, a home exercise program consisting of either muscle endurance training (i.e., 3 sets of 20 reps, 3 times a week for 12 months) or muscle strength training (i.e., 1 set of 15 reps, 3 times a week for 12 months) was found to be effective in improving function in the short-term.	Systematic review
Some evidence	Evidence statement	Design
	An exercise program that includes resistance training of the cervical and scapulothoracic muscles, combined with stretching of the same muscles, is likely to be beneficial for mechanical neck pain.	Meta analysis
	General fitness exercises and upper extremity exercises are unlikely by themselves to be beneficial for mechanical neck pain.	Meta analysis
	In patients with chronic mechanical neck pain, stabilization exercises with or without connective tissue massage are equally effective for reducing pain, anxiety and physical health while increasing the quality of life.	RCT

	A 3-week program of twice weekly home neck exercises with manual physical therapy that includes joint mobilization, muscle energy, and stretching, reduces neck pain and disability compared with a minimal intervention for patients with chronic neck pain at 6 weeks follow-up, but it did not persist at 1 year follow-up.	RCT
	In the setting of nonspecific acute or subacute neck pain that limits daily activities but is unaccompanied by radicular signs, weakness, or numbness in the arm, 3 treatment options include spinal manipulative therapy, prescription or nonprescription medication, and home exercise with advice. 12 weeks of spinal manipulative therapy is likely to be marginally more beneficial than medication, but is not more beneficial than home exercise with advice consisting of 2 hours of instruction by a physical therapist, focused on simple self-mobilization exercise of the neck and shoulder joints done 6 to 8 times per day.	RCT
	A combination of exercise and spinal manipulation is more effective than manipulation alone in relieving chronic neck pain, and these advantages remain for more than 1 year after the end of treatment.	RCT
	Multimodal care consisting of a 3-week program of twice weekly exercise with manual therapy with joint mobilization, muscle energy techniques, and home strengthening and ROM exercises, reduces neck pain and disability compared with a minimal intervention consisting of postural advice, ROM exercise, subtherapeutic ultrasound, and continuation of prescribed medication.	Systematic review
	A program of 2 sessions of thoracic thrust manipulation followed by a cervical exercise program is more effective than a cervical exercise program alone.	Systematic review
	Mobilization, manipulation, and exercise does not provide greater long-term pain relief when compared to exercise alone.	Systematic review
	In patients with tension-type musculoskeletal neck pain with no neurological deficits, both individually applied physical therapy and a 7-week group course teaching relaxation and body awareness activities lead to improvements in pain and function after 20 weeks. However, the benefits are likely to be greater in patients who receive the group instruction in relaxation and body awareness exercises. A brief course in this intervention is needed for the physical therapists who will implement the intervention at their clinics.	Systematic review

Table 11.

Evidence Table: The McKenzie Method and Neck Pain

Some evidence	Evidence statement	Design
	The McKenzie approach provides similar outcomes in improving pain, disability, and ability to carry out work activities in comparison with cognitive behavioral therapy (CBT).	RCT

Table 12.

Evidence Table: Pilates and Tai Chi in Neck Pain

Good evidence	Evidence statement	Design
	Pilates exercise shows statistically and clinically significant reductions in pain and statistically significant improvements in function in the short-term (maximum follow up 15 weeks) compared with usual care, no exercise, education, or physical activity for the treatment of patients with chronic low back pain.	Systematic review
Some evidence	Evidence statement	Design
	12 weeks of Tai Chi is more effective than no treatment to improve pain, disability, quality of life, and postural control in subjects with chronic nonspecific neck pain.	RCT

Table 13.

Evidence Table: Craniosacral Therapy and Neck Pain

Some evidence	Evidence statement	Design
	Craniosacral therapy for chronic nonspecific neck pain, performed by a physical therapist trained in the technique, is superior to sham treatment in reducing neck pain intensity at 8 weeks and probably at 20 weeks.	RCT

Table 14.

Evidence Table: Active Therapies with Education for Neck Pain

Some evidence	Evidence statement	Design
	Among patients with persistent grade I-II whiplash-associated disorders, the addition of a 10-week standardized progressive goal attainment program to a functional restoration physical therapy (2.5 hours/day, 3 times/week for 10 weeks or until return to work) improves the rate at which patients return to work.	Systematic review
	A 2-day course focusing on the biopsychosocial model with an emphasis on the goals of returning to usual activities and fitness is as effective in reducing disability as 6 sessions of manual therapy sessions provided by physiotherapists and more limited patient education.	RCT

Table 15.

Evidence Table: Pain Neuroscience Education

Summary:

For patients with chronic musculoskeletal pain, including neck pain, active therapy with pain neuroscience education reduces pain and increases function as compared to active therapy alone.

Good evidence	Evidence statement	Design
	Pain neuroscience education combined with a physical intervention is more effective in reducing pain, improving disability, and reducing healthcare utilization compared with either usual care, exercise, other education, or another control group for the treatment of patients with chronic musculoskeletal pain.	Systematic review
Some evidence	Evidence statement	Design
	The difference between the pain-education and control group for patients with chronic neck or low back pain reaches clinical relevance for both physical and mental-health domains in the 36-item Short Form Health Survey (SF-36 scale) and persists at 12 months.	RCT

Section 7.a.ii. Behavioral and Psychological Interventions

Introduction. Psychological therapeutic and diagnostic interventions have selected use in acute pain problems and more widespread use in sub-acute and chronic pain populations. Psychosocial interventions include psychotherapeutic treatments for mental health conditions, as well behavioral medicine treatments. Therapeutic psychological interventions include, but are not limited to, individual counseling and group therapy. Treatment can occur within an individualized model, a multi-disciplinary model, or a structured pain management program.

These interventions may similarly benefit patients without psychiatric conditions but who may need to make major life changes in order to cope with pain or adjust to disability. Health behavior assessment and intervention services are used to identify and address the psychological, behavioral, emotional, cognitive, and interpersonal factors important to the assessment, treatment, or management of physical health problems.

The following commonly used terminology appears in the context of addressing behavioral and psychological interventions:

- biofeedback training, including:
 - electromyogram,
 - skin temperature,
 - respiration feedback,
 - respiratory sinus arrhythmia,
 - heart rate variability,
 - electrodermal response, and
 - electroencephalograph,
- cognitive behavioral therapy (CBT);
- mindfulness-based stress reduction;
- progressive relaxation;
- relaxation therapy; and
- sleep hygiene training.

CBT is a psychological therapy that integrates theories of cognition and learning with treatment techniques derived from cognitive therapy and behavior therapy. It assumes that cognitive, emotional, and behavioral variables are functionally interrelated. Treatment is aimed at identifying and modifying the patient's unproductive thought processes and behaviors through cognitive restructuring and behavioral techniques to achieve change. Variations of CBT methods can be used to treat a variety of conditions, including chronic pain, depression, anxiety, phobias, and post-traumatic stress disorder. For patients with multiple diagnoses, more than 1 type of CBT might be needed. A "manualized" approach to CBT follows a specific protocol in a manual. See the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) for additional information on psychological evaluation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Behavioral and Psychological Interventions.

- Contraindications include active suicidality, homicidality, active psychosis, or major untreated psychological comorbidity.
- Relative contraindication includes a lack of patient engagement despite interventions targeting initial resistance.
- Relative contraindications to CBT include literacy below the 6th grade level, lack of English language proficiency, and cognitive impairment.

Side Effects and Adverse Events Related to Behavioral and Psychological Interventions.

- Side effects and adverse events include emotional discomfort, cognitive dissonance, irritability, interpersonal difficulties, or temporary increase in stress.

Recommendations.

Core Requirements.

Recommendation 54. Formal psychological or psychosocial evaluation is recommended as a component of the biopsychosocial approach to patient recovery from physical injury and should attempt to identify both primary psychological risk factors (e.g., psychosis and active suicidality) and secondary psychological risk factors (e.g., moderate depression and job dissatisfaction) (table 16). The evaluation should distinguish between pre-existing, aggravated, and/or purely causative psychological conditions.

Recommendation 55. Comprehensive psychological evaluation must be performed by a psychologist with PhD, PsyD, or EdD credentials or a physician with psychiatric MD/DO credentials.

Recommendation 56. Psychometric testing, distinct from screening psychological questionnaires, must be administered by a psychologist with a PhD, PsyD, or EdD or a health professional working under the supervision of a doctorate level psychologist. A physician with appropriate training may also administer such testing, but interpretation of the tests should be done by a properly credentialed mental health professional.

Recommendation 57. Evaluation for psychiatric medication is permitted if there is an established diagnosis consistent with the standards of the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM). See the Medications section.

Recommendation 58. An initial assessment, ongoing assessment, and treatment plan with measurable and clinically meaningful behavioral goals, time frames, and specific planned interventions are required before initiating psychological/psychiatric interventions. Ongoing requirements include:

- a biweekly status report during initial, more frequent treatment and monthly thereafter, documenting progress toward functional recovery;

- a discussion of the psychosocial issues affecting the patient's ability to participate in treatment; and
- addressing pertinent issues such as pre-existing, aggravated, and/or causative issues, as well as realistic functional prognosis.
- Recommendation 59. The following is recommended when obtaining a psychological evaluation:
- a full release from the patient prior to evaluation acknowledging that information that may go to the employer, and
- exclusion of sensitive health information not directly related to the work-related conditions in reports sent to the insurer is essential.

CBT Requirements.

Recommendation 60. CBT is recommended for cervical spine pain patients who catastrophize, cope ineffectively with pain, or avoid activity out of fear of re-injury ([table 17](#)).

Recommendation 61. A full psychological evaluation is required before CBT can be initiated.

Recommendation 62. A CBT provider must be a:

- psychologist with PhD, PsyD, or EdD credentials;
- psychiatric MD/DO; or
- licensed mental health or licensed health care provider with training in CBT or certified as a CBT therapist with experience in treating chronic pain disorders who works in consultation with a psychologist with a PhD, PsyD, EdD, or psychiatric MD/DO.

Recommendation 63. A manualized approach to CBT is strongly recommended if being performed by a non-mental health professional, with appropriate supervision by a psychologist with a PhD, PsyD, EdD, or psychiatric MD/DO provider.

Recommendation 64. Candidates for CBT will have all of the following characteristics:

- adequate literacy level to complete homework used to teach inductive rational thinking,
- adequate cognitive and education abilities to meet the requirements of a CBT protocol, and
- otherwise stable social circumstances.

The selection of CBT methods should be based on the individual's literacy level, English proficiency, and cognitive capabilities, as assessed by the behavioral health provider. Individuals who are not candidates for CBT may benefit from other behavioral and psychological approaches.

Biofeedback Requirements.

Recommendation 65. Biofeedback as an adjunct to psychological therapy is recommended for patients who meet 1 of the following indications:

- musculoskeletal injury, in which muscle dysfunction or other physiological indicators of excessive prolonged stress response affects and/or delays recovery;
- a need for training to improve self-management of pain, anxiety, panic, anger or emotional distress, opioid withdrawal, insomnia, sleep disturbance, and/or other central and autonomic nervous system imbalances; and/or
- motivation to learn and practice biofeedback and self-regulation techniques ([table 18](#)).

Recommendation 66. If biofeedback treatment is indicated, it must be done in conjunction with the patient's other psychosocial or medical interventions.

Recommendation 67. Psychologists or psychiatrists who provide psycho-physiological therapy, which integrates biofeedback with psychotherapy, must be either Biofeedback Certification International Alliance (BCIA) certified or practicing within the scope of their training.

- Non-licensed health care providers of biofeedback for chronic pain patients must be BCIA certified and shall have their biofeedback treatment plan approved by the authorized treating psychologist or psychiatrist.
- Biofeedback can also be provided by health care providers who follow a set treatment and educational protocol. Such treatment may utilize standardized material, relaxation recordings, or app-based interventions. App-based interventions must meet the criteria in Rule 18.

Substance Use Disorder Requirements.

Recommendation 68. Patients with substance use disorder, high-dose opioid use, or use of other drugs of abuse may require inpatient and/or outpatient chemical dependency treatment programs before or in conjunction with other interdisciplinary rehabilitation. See the Chronic Pain Disorder Medical Treatment Guidelines (MTGs).

Time Frames.

Time Frames for Behavioral and Psychological Interventions			
	Time to produce effect (sessions)	Frequency (sessions/week)	Maximum duration
Group CBT	up to 8 (2-hours)	up to 2	16 sessions
Individual CBT	up to 8 (1-hour)	up to 2	16 sessions
Biofeedback	up to 4 sessions	up to 2	12 sessions [¶]
Relaxation	up to 4 sessions	up to 2	12 sessions [¶]
Other psychological interventions	up to 8	up to 2 [*]	6 months [‡]
[¶] Treatment beyond 12 sessions must be documented with respect to need, expectation, and ability to facilitate positive symptomatic and functional gains. [*] For the first 2 weeks of treatment, excluding hospitalization, if required; decreasing to weekly in the 2nd month of treatment and then 2-4 times per month (with the exception of exacerbations, which may require increased frequency of visits). [‡] Longer supervised psychological/psychiatric treatment may be required, especially if there are ongoing medical procedures or complications. If counseling beyond 6 months is indicated, the management of psychosocial risks or functional progress must be documented.			

Evidence Tables.

Table 16.		
<i>Evidence Table: Psychological Assessments and Outcomes</i>		
Summary: Psychological testing identifies patients most likely to benefit from surgery.		
Good evidence	Evidence statement	Design
	Psychometric testing can predict medical treatment outcomes.	Cohort study
	Interdisciplinary programs that include screening for psychological issues, identification of fear-avoidance beliefs and treatment barriers, and establishment of individual functional and work goals will improve function and decrease disability.	RCT
Some evidence	Evidence statement	Design
	Psychological and medical risk factor assessment prior to surgery can identify patients unlikely to benefit from surgery.	Cohort study

Table 17.		
<i>Evidence Table: CBT</i>		
Summary: CBT may reduce pain, disability, sick leave, health care utilization, and insomnia for patients with chronic pain.		
Good evidence	Evidence statement	Design
	CBT may reduce pain and disability in patients with chronic pain, but the magnitude of the benefit is uncertain.	Meta analysis
	CBT, but not behavioral therapy, shows weak to small effects in reducing pain, and small effects on improving disability, mood, and catastrophizing in the treatment of patients with chronic pain.	Meta analysis
	A stepped care program is more effective than usual care in veterans with chronic musculoskeletal pain. The stepped care program consisted of 12 weeks during which nurse case managers took a medication use history and adjusted medication dosage and scheduling through telephone contacts with patients every other week, followed by a 12-week step in which cognitive behavioral treatment was administered by 45 minute individual sessions by telephone every other week. Disability and pain interference with daily activity with stepped care were both superior to usual care in which patients were given printed handouts and were followed for all care by their primary treating physicians.	Meta analysis

Some evidence	Evidence statement	Design
	A 6-week program of cognitive-behavioral group intervention with or without physical therapy can reduce sick leave, health care utilization, and the risk for developing long-term sick leave disability (> 15 days) in workers with nonspecific low back or neck pain compared with simple verbal instruction by a physician.	RCT
	CBT provided in seven 2-hour small group sessions can reduce the severity of insomnia in chronic pain patients.	RCT
	The McKenzie approach provides similar outcomes in improving pain, disability, and ability to carry out work activities in comparison with CBT.	RCT
	The benefits of CBT continue to be present 3 or more years after the initial intervention, although the differences between CBT and the control group are likely to become narrower with extended follow-up.	RCT
	Behavioral modification, such as patient education and group or individual counseling with CBT, can be effective in reversing the effects of insomnia.	RCT

Table 18.

Evidence Table: Biofeedback and Relaxation Training

Summary:

Both biofeedback and relaxation therapy may reduce pain and disability for patients with chronic low back pain. While the literature in this table discusses chronic low back pain, there is presumed physiologic similarity for chronic neck pain.

Good evidence	Evidence statement	Design
	Biofeedback or relaxation therapy is equal in effect to CBT for chronic low back pain.	Meta analysis
	Self-regulatory interventions, such as biofeedback and relaxation training, may be equally effective.	Meta analysis
Some evidence	Evidence statement	Design
	Progressive relaxation was the only psychological therapy associated with improvement in function as well as pain intensity compared to wait-list control in chronic low back pain.	Systematic review

Section 7.b. Adjunct Second Line Treatments, as Indicated

Section 7.b.i. Passive Therapies

Introduction. Passive therapies include treatments that do not require a patient's energy expenditure. They are most helpful early in treatment, and they are directed at controlling symptoms and improving the rate of healing soft tissue injuries. They can be used adjunctively with active therapies to help control swelling, pain, and inflammation. The following passive therapies are common treatments: treatments:

- acupuncture,
- acupuncture with electrical stimulation,

- iontophoresis,
- joint mobilization,
- low level laser,
- manipulation,
- manual treatment,
- manual traction,
- massage,
- mechanical or motorized traction,
- phonophoresis,
- soft tissue mobilization,
- short-wave diathermy,
- superficial heat and cold therapy,
- trigger point dry needling,
- ultrasound, and
- unattended electrical stimulation (e.g., transcutaneous electrical nerve stimulation [TENS]).

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Passive Therapies.

- High velocity / low amplitude manipulation is contraindicated in those with joint instability, fractures, severe osteoporosis, infection, metastatic cancer, active inflammatory arthritis, signs of progressive neurologic deficits, myelopathy, vertebrobasilar insufficiency, or carotid artery disease.
- Relative contraindications to manipulation include spinal stenosis, spondylosis, and disc herniation.
- Manual traction is contraindicated in those with tumor, infection, fracture, or fracture dislocation.
- Non-oscillating inversion traction methods are contraindicated in patients with glaucoma or hypertension.

Complications of Passive Therapies.

- Rare complications of dry needling or acupuncture include pneumothorax, infection, or syncope.
- Cervical vertebral artery dissection can occur with high velocity neck manipulation.

Side Effects and Adverse Events Related to Passive Therapies.

- Acupuncture may result in treatment-related pain or bruising associated with needle insertion.

Recommendations.

Core Requirements.

Recommendation 69. Patients in passive therapy must demonstrate functional progress through validated functional assessment measures. If there is no evidence of functional progress within the time to produce effect, the therapy shall be discontinued and the patient must be referred back to their treating provider for evaluation. Each patient is limited to a maximum of 4 discrete passive therapy trials.

Recommendation 70. Passive therapies must occur concurrently with self-directed exercise or formal active therapy programs (table 19).

Recommendation 71. The frequency of passive therapy must decrease over time.

Recommendation 72. Durations of care beyond those listed as “time to produce effect” and “maximum” are warranted in certain circumstances when treatment to date has resulted in measurable and clinically meaningful functional improvement. These circumstances include:

- re-injury;
- interrupted continuity of care;
- after surgery, particularly multiple surgeries;
- injuries (e.g., fracture); or
- comorbidities.

Specific goals with objective measures of functional improvement must be cited to justify extended durations of care.

Recommendation 73. Manipulation under general anesthesia and manipulation under joint anesthesia are not recommended.

Recommendation 74. Mechanical traction is not generally recommended for neck pain patients with or without radicular symptoms.

Recommendation 75. Due to the absence of quality evidence supporting their use, ultrasound, phonophoresis, low level laser, kinesiotaping, iontophoresis, and motorized traction are not recommended for neck pain.

Time Frames.

Time Frames for Passive Therapies			
	Time to produce effect (sessions)	Frequency* (sessions/week)	Maximum duration
Mobilization and Manipulation	up to 6	up to 3 times/week	8 weeks
Massage	1	up to 2 times/week	8 weeks
Acupuncture	up to 6	up to 3 times/week	15 treatments
Heat/cold, short-wave diathermy, unattended electrical stimulation (e.g., TENS**)	up to 4	up to 3 times/week	8 weeks
Trigger point / dry needling	up to 4	up to 2 times/week	8 weeks
Traction (manual)***	up to 3	up to 3 times/week	4 weeks
*See recommendation 71 regarding the expected decreasing frequency over time. **If TENS treatment results in documented functional benefit and is anticipated to extend beyond 4 treatments, consider purchase of a home TENS unit. ***If response is negative after 3 thirty minute treatments, discontinue.			

Evidence Tables.

Table 19.

Evidence Summary: Passive Therapies

Passive therapies may provide functional benefit and symptomatic relief in patients with neck pain, especially when paired with an active intervention. The decision to refer a patient for 1 passive therapy over another should be based on patient preference and relative safety, not on the expectation of a greater treatment effect among passive therapy options.

Table 20.

Evidence Table: Manipulation and Mobilization for Neck Pain

Summary:

Manipulation and/or mobilization can reduce pain and improve function for patients with neck pain, especially when combined with exercise.

Good evidence	Evidence statement	Design
	Manipulation alone or mobilization alone provides immediate, short-term, and intermediate-term relief for acute, subacute, and chronic neck pain.	Meta analysis
	Adding exercise in combination with other interventions such as: 1) manipulation alone, or 2) manipulation and mobilization, or 3) mobilization, muscle energy, and stretching, is more effective than manipulation alone, mobilization alone, exercise alone, and other minimal intervention or education alone in reducing neck pain and disability.	Systematic review
	In patients with neck pain (with or without radicular findings), multiple sessions of thoracic manipulation was more effective in reducing intermediate-term chronic neck pain when compared with multiple sessions of a control.	Meta analysis
	In people with chronic neck pain, exercise with or without manipulation results in decreased pain and disability compared to home exercise and advice at 12 weeks but not 52 weeks. The percentage of patients with 2.5% decrease in pain at 1 year was 51% with manipulation and exercise, 57% with exercise only, and 41% with home exercise and advice. It is not clear that the expense is worth the outcome long-term.	RCT
	There is little difference in clinically significant benefit between high-velocity, low-amplitude cervical thrust manipulation and low velocity intervention, like mobilization, using techniques such as muscle energy, myofascial release, and sustained apophyseal natural glides.	Systematic review

Table 20 continued.

Evidence Table: Manipulation and Mobilization for Neck Pain

Good evidence cont.	In the setting of chronic neck pain, an intervention that mobilizes a nearby but asymptomatic spinal segment may be as beneficial as an intervention done directly at the symptomatic level. This may give the clinician additional options	Systematic review
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	regarding the optimal treatment technique to be applied to the patient.	
Some evidence	Evidence statement	Design
	A 3-week program of twice weekly home neck exercises with manual physical therapy that includes joint mobilization, muscle energy, and stretching, reduces neck pain and disability compared with a minimal intervention for patients with chronic neck pain at 6 weeks follow-up, but it did not persist at 1 year follow-up.	RCT
	Mobilization, manipulation, and exercise does not provide greater long-term pain relief when compared to exercise alone.	Systematic review
	Thoracic thrust manipulation may improve pain and function for mechanical neck pain.	Meta analysis
	A combination of exercise and spinal manipulation is more effective than manipulation alone in relieving chronic neck pain, and these advantages remain for more than 1 year after the end of treatment.	RCT
	Multimodal care consisting of a 3-week program of twice weekly exercise with manual therapy with joint mobilization, muscle energy techniques, and home strengthening and ROM exercises, reduces neck pain and disability compared with a minimal intervention consisting of postural advice, ROM exercise, subtherapeutic ultrasound, and continuation of prescribed medication.	Systematic review
	A program of 2 sessions of thoracic thrust manipulation followed by a cervical exercise program is more effective than a cervical exercise program alone.	Systematic review
	In the setting of nonspecific acute or subacute neck pain that limits daily activities but is unaccompanied by radicular signs, weakness, or numbness in the arm, 3 treatment options include spinal manipulative therapy, prescription or nonprescription medication, and home exercise with advice. 12 weeks of spinal manipulative therapy is likely to be marginally more beneficial than medication, but is not more beneficial than home exercise with advice consisting of 2 hours of instruction by a physical therapist, focused on simple self-mobilization exercise of the neck and shoulder joints done 6 to 8 times per day.	RCT

Table 21.

Evidence Table: Massage for Neck Pain

Some evidence	Evidence statement	Design
	In patients with chronic mechanical neck pain, stabilization exercises with or without connective tissue massage are equally effective for reducing pain, anxiety and physical health while increasing the quality of life.	RCT

Table 22.		
<i>Evidence Table: Trigger Point Dry Needling and Neck Pain</i>		
Some evidence	Evidence statement	Design
	4 sessions of trigger point deep dry needling with passive stretching over 2 weeks was significantly more effective in reducing neck pain and improving neck disability than passive stretching alone in the short-term and at 6-month follow-up in people with chronic nonspecific neck pain.	RCT

Table 23.		
<i>Evidence Table: Acupuncture</i>		
Summary: Acupuncture may reduce pain and improve function for patients with chronic neck pain.		
Good evidence	Evidence statement	Design
	Acupuncture is useful in the treatment of neck pain.	Systematic review
Some evidence	Acupuncturists must be properly trained in aseptic technique.	Systematic review
	Evidence statement	Design
	Patients with nonspecific chronic neck pain were randomly assigned to 12 acupuncture sessions or 20 one-to-one Alexander Technique lessons with equivalent overall contact time. Both were associated with a significant reduction in neck pain and decreased disability, per the Northwick Park Neck Pain Questionnaire (NPQ), at 12 months compared with usual care.	RCT
Some evidence	In the current study, participants with chronic neck pain received a 3-week phase of acupuncture, which included a total of 6 treatments, 3 times per week, each lasting 30 minutes. It was found that up to 12 weeks after treatment ended, those who received classic acupuncture improved significantly more with respect to disability and pain than those who received acupuncture at non-classical sites. Both groups had significant improvement.	RCT

Table 24.		
<i>Evidence Table: Acustimulation and Neck Pain</i>		
Good evidence	Evidence statement	Design
	There is no difference between the use of electroacupuncture and sham laser acupuncture on chronic neck pain at 3 months after 9 treatments and 6 months, although both groups improved.	RCT

Some evidence	Evidence statement	Design
	A combination of electrical “acustimulation” to the wrist combined with neck stretching and strengthening exercises for 30 minutes 2 times per week for a period of about 4 weeks demonstrates more improvement in chronic neck pain and patient self-confidence in performing functional activities than neck exercises alone for up to 1 month.	RCT

Table 25.

Evidence Table: Traction and Neck Pain

Some evidence	Evidence statement	Design
	Intermittent cervical traction does not add therapeutic benefit to a brief course of individualized manual therapy combined with exercise for patients with cervical radiculopathy.	RCT

Section 7.b.ii. Durable Medical Equipment

Introduction. Durable medical equipment includes devices that can be used repeatedly to serve a medical purpose. These include braces or splints for supporting, immobilizing, or treating muscles, joint, or skeletal parts that are weak, ineffective, deformed, or injured.

Complications / Side Effects and Adverse Events.

Complications of Durable Medical Equipment.

- Complications of durable medical equipment include deconditioning of the cervical musculature and associated fatigue or pain, and skin irritation resulting from contact with the orthotic.

Recommendations.

Acute and Postoperative Treatment Time Frame Requirements.

Recommendation 76. Soft cervical collars are not recommended for an isolated cervical sprain ([table 26](#)).

Recommendation 77. Rigid cervical collars are permitted in the postoperative and emergency setting. Documentation should include directions for duration and frequency of use.

Recommendation 78. Halo immobilization is recommended in the treatment of cervical fracture, dislocation, and instability, at the discretion of the treating surgeon.

Evidence Table.

Table 26.		
<i>Evidence Table: Cervical Collars for Neck Injury</i>		
Some evidence	Evidence statement	Design
	Patients encouraged to continue usual activity have less neck stiffness and headache than patients placed in cervical collars and placed on sick leave following motor vehicle crashes.	RCT
	Semi-hard collars worn during the day for 3 weeks and then weaned over 3 weeks may hasten resolution of recent onset cervical radiculopathy.	RCT

Section 8. Third Line Treatment, as Indicated

Section 8.a. Injections - Diagnostic and Therapeutic

Introduction. Spinal injections are considered when function remains limited despite appropriate non-operative treatment options. Diagnostic spinal injections are used to localize a source of pain when the results would impact the course of treatment. Therapeutic injections are used to temporarily reduce functional impairment. Most individuals with neck pain will not need a spinal injection. See the Overview of Care section.

Contraindications / Complications / Side Effects and Adverse Events.

- See the specific injection section.

Core Requirements.

Recommendation 79. Spinal injections are indicated when all of the following criteria are met:

- positive correlation among clinical findings, the clinical course, and diagnostic tests;
- positive functional response to a diagnostic injection, if required;
- persistent functional impairment despite engagement in 6 weeks of active therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

Recommendation 80. All spinal injections should be preceded by magnetic resonance imaging (MRI) or computed tomography (CT) scan.

Recommendation 81. Informed consent is required before all invasive procedures. See the Essential First Line Treatment section.

Recommendation 82. Routine use of sedation is not recommended for diagnostic or therapeutic spinal injections. Minimal to moderate sedation and pain relief may be needed for some patients but are only permitted when there is documentation of extreme circumstances.

Recommendation 83. All injections must be accompanied by self-directed exercise or formal physical therapy.

Recommendation 84. Multi-planar fluoroscopic imaging is required to create permanent images that confirm proper needle placement and contrast flow pattern. CT needle guidance may be required in rare circumstances.

Recommendation 85. Total steroid injections at all sites, including the extremities, should be limited to 4 per year to avoid side effects from steroids. Prior authorization is required for additional injections with appropriate documentation of medical reasoning and functional improvement.

Recommendation 86. Due to the absence of quality evidence supporting their use, the injection of substances other than steroids, anesthetic, or contrast solution is not recommended (e.g., orthobiologics such as platelet rich plasma, stem cells, or amniotic fluid; ozone; etc.).

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.a.i. Epidural Steroid Injection (ESI) and Transforaminal Nerve Root Block

Introduction. Epidural steroid injections (ESIs) are corticosteroid injections into the cervical epidural space by a transforaminal or interlaminar approach. A transforaminal nerve root block involves injecting steroid medication near where the nerve root exits the spinal column. The therapeutic goals of injections are to restore ROM and facilitate progress in active therapy programs by temporarily reducing pain and inflammation in the acute or subacute phases of injury.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to ESI.

- Poorly controlled diabetes mellitus, systemic or localized bacterial infection, known anatomic abnormalities, and bleeding diathesis are absolute contraindications to ESI.
- Patients with existing osteoporosis or other risk factors for osteoporosis should rarely receive ESIs.
- Patients on medications that affect coagulation require special consideration prior to ESI.
- Allergy to contrast material, somatization disorder, poorly controlled congestive heart failure, and uncontrolled hypertension are relative contraindications to ESI.
- Significant spinal stenosis is a contraindication to an injection at that level. This applies to placing a needle at the spinal stenosis area, but it is acceptable to inject adjacent to the level.

Complications of ESI.

- Long-term morbidity can occur as a result of ESI, although complications that lead to long-term morbidity are rare.
- Permanent paresis, anaphylaxis, and arachnoiditis have been rarely reported with the use of epidural steroids.
- Cervical epidural injections carry additional risks of injury including death, spinal cord injury, and stroke when compared to lumbar injections.
- Headache, nerve injury, transient neurapraxia, vasovagal effects, infection, local pain, epidural hematoma, spinal cord injury, dural perforation, and cerebrospinal fluid leaks have been reported.
- Direct needle trauma is a reported complication (recommendation 90).
- Although major neurological complications are rare with fluoroscopically-guided ESIs, there is a much greater risk with injections above C5-C6 and with particulate steroid injections (recommendation 91; table 28).
- Osteoporotic fractures, Cushing's syndrome, hypopituitarism, spinal epidural lipomatosis, and growth hormone deficiency have been associated with ESI.

Side Effects and Adverse Events Related to Injectate.

- Increased blood glucose levels, decreased plasma cortisol levels, and impaired bowel or bladder function are possible side effects of ESI.
- Dose-dependent suppression of the hypothalamic-pituitary-adrenal axis, leading to a transient secondary adrenal insufficiency in the first few weeks after ESI (table 27).

Recommendations.

Core Requirements.

Recommendation 87. Interlaminar injections should not be performed above the level of C6-C7 nor at the level of any stenosis as demonstrated on pre-procedure imaging review.

Recommendation 88. For a patient to be a candidate for an ESI, the provider must document all of the following:

- subjective report of severe radicular pain that correlates with objective findings;
- positive neural compression test (e.g., dural tension, Spurling's sign, traction/distraction, upper limb tension test) and/or reflex, motor, or sensory changes on examination specifically correlating with imaging findings; and
- imaging findings that demonstrate impingement of nerve(s) or spinal cord.

Recommendation 89. ESI is not recommended for the treatment of axial neck pain in the absence of radicular symptoms.

Recommendation 90. Multi-planar fluoroscopic imaging is required to create permanent images confirming proper needle placement and contrast flow pattern.

Recommendation 91. Particulate corticosteroid preparations are not recommended for epidural injection ([table 28](#)).

Recommendation 92. A patient who meets the ESI indications can trial an epidural injection of local anesthetic alone as an alternative to an injection containing steroid ([table 29](#)).

Recommendation 93. All patients with diabetes must be instructed to monitor their glucose levels carefully over the 7 days after steroid injection. Clinicians can pursue diabetic screening tests for those who appear to be at risk for Type 2 diabetes ([table 30](#)).

Recommendation 94. A diagnostic response to a selective nerve root block must be documented to show its value, including:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; tolerance and time limits for reaching and lifting); and
- at least 80% radicular pain improvement as measured using an accepted pain scale (e.g., visual analog scale [VAS] or numeric rating scale [NRS]) that is consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings and preferably for 1 week if local anesthetic is used. Documentation of pain response may be indicated for up to 3 weeks, depending on the nature of the injection.

A successful response to a diagnostic injection requires documentation of positive functional changes by trained personnel and may include nurses, physician assistants, medical assistants, therapists, or non-injectionist physicians. Functional progress supersedes pain improvement.

Acute Phase Requirements (Within 6 Weeks of Initial Care).

Recommendation 95. Early referral within 6 weeks for ESI is permitted when the provider documents all criteria in recommendation 88 and at least 1 of the following:

- radicular pain-related functional impairment prevents participation in active therapy and activities of daily living (ADLs); or
- an opioid treatment is being considered.

Subacute and Chronic Phase Requirements (After 6 Weeks of Care).

Recommendation 96. ESI or referral for surgical consultation is permitted for patients who meet the criteria in recommendation 88 and both of the following:

- functional impairment despite 6 weeks of optimal medical management and participation in active therapy, and
- confounding psychosocial risk factors have been screened for and are being clinically addressed. See the Behavioral and Psychological Interventions section (table 31).

Repeat ESI Requirements.

Recommendation 97. Repeat ESI can occur as soon as 2 weeks after the first injection when the initial injection resulted in a sustained therapeutic response for 2 weeks or more and is characterized by the following:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; tolerance and time limits for reaching and lifting), and
- 80% radicular pain improvement as measured using an accepted pain scale (e.g., VAS or NRS).

Functional progress supersedes pain improvement. The third injection can then be repeated after a hiatus of 6 months ([Appendix Functional Screens](#)). See the [Chronic Pain Disorder Medical Treatment Guidelines \(MTGs\)](#) for information on subsequent injections.

Recommendation 98. Ordering a “series” of ESIs is prohibited. If the first injection fails to result in functional gain and pain relief as outlined in [recommendation 97](#), the injection cannot be repeated at that level with the same approach.

Recommendation 99. If the decision is made to proceed with a different type of injection, the specific criteria for that injection must be satisfied.

Recommendation 100. Testing morning cortisol levels is recommended prior to the third or fourth steroid injection.

Evidence Tables.

Table 27.		
<i>Evidence Table: ESI and the Hypothalamic-Pituitary-Adrenal Axis</i>		
Some evidence	Evidence statement	Design
	In the setting of ESI given for low back pain with radiculopathy, there is a dose-dependent suppression of the hypothalamic-pituitary-adrenal axis leading to a transient secondary adrenal insufficiency in the first few weeks after the procedure, with an 80 mg dose of methylprednisolone acetate leading to secondary adrenal insufficiency in about 80% of patients the first week after the injection, and in about 50% of patients receiving a 40 mg dose. The cortisol response to an injection of adrenocorticotrophic hormone (ACTH) gradually returns to normal in the weeks after the injection, but secondary adrenal insufficiency remains detectable in some patients 4 weeks later.	RCT

Table 28.

Evidence Table: Risk of particulate ESI

Summary:

ESIs with particulate steroids, such as methylprednisolone and triamcinolone, have systemic absorption effects and a much higher risk of major neurological complications than ESIs of water soluble steroids, such as betamethasone and dexamethasone.

Strong evidence	Evidence statement	Design
	Although major neurological complications are rare with fluoroscopically-guided cervical and lumbar ESIs, the risk is much greater with particulate steroid injections.	Systematic review
Good evidence	Evidence statement	Design
	Some steroid preparations have a systemic absorption effect, which is reflected in suppression of morning cortisol measurements 3 and 6 weeks after the injection. This appears to be the case with the particulate steroids, methylprednisolone and triamcinolone, but not with the water soluble steroids, betamethasone and dexamethasone.	RCT
	Particulate steroids offer no advantage over nonparticulate steroids in the setting of transforaminal or interlaminar ESIs for patients with lumbar or cervical radiculopathy, either with respect to pain relief or improvement of function.	Systematic review

Table 29.

Evidence Table: Use of Steroid in Epidural Injections

Summary:

The addition of a corticosteroid may not provide added improvement in function or pain over epidural injections with local anesthetic or saline alone. While the evidence does not support a benefit of adding steroids to local anesthetic spinal injections, steroids are routinely used in therapeutic spinal injections due to a presumed physiologic effect.

Some evidence	Evidence statement	Design
	Epidural injections of solutions not containing steroids (such as normal saline or local anesthetic in saline) are likely to have therapeutic effects in the setting of radicular pain, possibly arising from lavage effects and other effects not depending on the steroid component of the injected solution. The interpretation of clinical trials in which epidural steroid and non-steroid injections appear to have similar effects should be influenced by this principle.	Systematic review
	Results from transforaminal injections of anesthetic plus steroids is similar to injections with anesthetic only in the cervical spine.	RCT

Table 30.

Evidence Table: Corticosteroid-Containing ESI in Diabetic Patients

Good evidence	Evidence statement	Design
	In the setting of ESI for patients with adequately controlled diabetes mellitus, significant increases in fasting and postprandial blood glucose can be expected on the day of the procedure and for the first 3 days thereafter; however, the increase in blood glucose are less with a 20 mg than with a 40 mg dose of triamcinolone. There is no difference in pain relief between the 2 steroid doses, making the 20 mg dose preferable to a 40 mg dose.	RCT

Table 31.

Evidence Table: Depression and Cervical ESI

Some evidence	Evidence statement	Design
	Among patients with cervical radiculopathy who are considered to be surgical candidates, for whom a trial of non-surgical treatment is an option, the presence of clinical depression, defined as a score greater than 33 on the Zung Depression scale, is an unfavorable predictor for a successful response to a cervical ESI, reducing the favorable response rate by more than 50%.	Cohort study

Section 8.a.ii. Zygapophyseal (Facet) Injection

Introduction. Facet joint injection consists of the intra-articular or pericapsular injection of local anesthetic and corticosteroid. It has very limited uses.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Facet Joint Injection.

- Systemic or localized bacterial infection and bleeding diathesis are absolute contraindications to steroid injections. Poorly controlled diabetes is a contraindication if steroids are used.
- Patients with existing osteoporosis or other risk factors for osteoporosis should rarely receive steroid injections. It is unknown if facet joint injections contribute to increased risk of vertebral fractures, but appropriate precautions should be taken for at-risk patients.
- Patients on medications that affect coagulation require special consideration prior to these injections.
- Somatization disorder, poorly controlled congestive heart failure, and uncontrolled hypertension are relative contraindications to these injections. Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of Facet Joint Injection.

- Complications include nerve injury, bleeding, infection, contrast allergy, localized pain, or irritation.
- Extravasation of local anesthetic to the associated spinal nerve may result in associated weakness and numbness of that level.

Side Effects and Adverse Events Related to Facet Joint Injection.

- Diabetic patients may notice a steroid-induced effect on blood glucose levels in the post-injection period.

Recommendations.

Core Requirements.

Recommendation 101. Facet joint injections are of very limited use. One of the following requirements must be met prior to proceeding with a facet joint injection:

either

- at least 3 months of pain, unresponsive to 6 weeks of conservative therapies, including manual therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed; and
- physical examination findings are consistent with facet origin pain (e.g., pain on extension with lateral bending and referral patterns are consistent with the expected pathologic level) that is affecting activity;

or

- the patient has refused a rhizotomy despite facet origin pain on clinical exam;

or

- the patient has facet findings with a thoracic component.

Recommendation 102. Repeat facet joint injections are only recommended when the first injection results in a sustained therapeutic response for 3 months or more and is characterized by the following:

- improvement in at least 3 physical examination findings consistent with facet origin pain (e.g., pain on extension with lateral bending and referral patterns consistent with the expected pathologic level), and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS).

Functional progress supersedes pain improvement.

Recommendation 103. There is no indication for combined facet injection and medial branch block. Intraarticular facet joint steroid injections are not of diagnostic utility in determining appropriateness for radiofrequency (RF) neurotomy.

Recommendation 104. Facet injections cannot be performed at more than 2 levels, unilaterally or bilaterally.

Time Frames.

Time Frames for Facet Joint Injections	
Time to produce effect	Maximum duration
Approximately 30 minutes for local anesthetic; 48 to 72 hours for corticosteroid.	2 injections per level per year when at least 3 months of functional benefit is documented. Prior authorization must be obtained for injections beyond 2 levels.

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.a.iii. Intradiscal Injection

Introduction. Intradiscal steroid injection is a direct injection of a steroid-containing solution into an intervertebral disc.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Intradiscal Steroid Injections.

- Complications include infection, pain, bleeding, nerve injury, and allergic reaction to contrast dye.

Recommendations.

Core Requirements.

Recommendation 105. Intradiscal injections are not recommended ([table 32](#)).

Evidence Tables.

Table 32.		
Evidence Table: Intradiscal Steroid Injection		
Good evidence	Evidence statement	Design
	Intradiscal steroid injection is unlikely to relieve pain or provide functional benefit in patients with non-radicular back pain and therefore, they are not recommended.	RCT

Section 8.a.iv. Medial Branch Block and Radiofrequency (RF) Denervation

Introduction. Medial branch blocks are diagnostic injections used to determine whether a patient is a candidate for radiofrequency (RF) medial branch neurotomy. Medial branch neurotomy, also known as facet rhizotomy, is a procedure designed to denervate the facet joint by ablating the corresponding sensory medial branches. Continuous thermal percutaneous RF is the method frequently used. See the [Chronic Pain Disorder Medical Treatment Guidelines \(MTGs\)](#) for information on the use of repeat RF neurotomy in chronic pain maintenance management.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Medial Branch Block or RF Neurotomy.

- Systemic or localized bacterial infection and bleeding diathesis are absolute contraindications to medial branch blocks and RF neurotomy. Poorly controlled diabetes is a relative contraindication if steroids are used post-neurotomy.
- Allergy to contrast material may be a contraindication if alternative contrast material is unavailable.

Complications of Medial Branch Block or RF Neurotomy.

- Complications of medial branch block or RF neurotomy include bleeding, infection, neural injury, localized neuritis, or rarely, a deafferentation centralized pain syndrome.
- Complications of repeat neurotomy include atrophy of spinal musculature with repeat RF neurotomy.

Side Effects and Adverse Events Related to Medial Branch Block or RF Neurotomy.

- Side effects and adverse events related to medial branch block or RF neurotomy include localized pain at the injection and/or RF site and post-injection dysesthesia.

Recommendations.

Diagnostics Medial Branch Block Requirements.

Recommendation 106. Diagnostic medial branch blocks are recommended when all of the following are present:

- physical examination findings consistent with facet origin pain (e.g., pain on extension with lateral bending and referral patterns consistent with the expected pathologic level);
- at least 3 months of pain, unresponsive to 6 to 8 weeks of conservative therapies, including manual therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed. See the Behavioral and Psychological Interventions section.

Recommendation 107. Diagnostic medial branch blocks are limited to 2 anatomic facet joint levels or 3 medial branch levels. Bilateral controlled blocks are permitted, if performed in a way that preserves diagnostic accuracy.

Recommendation 108. A diagnostic response to medial branch blocks must be documented to show its value, including:

- improvement in at least 3 objective functional measures (e.g., spinal ROM; reaching and lifting); and
- at least 80% improvement in an accepted pain scale (e.g., VAS or NRS) reported with post-injection provocative testing that is consistent with:
 - the expected duration of the injected local anesthetic phase, and
 - a post-injection pain diary with at least 8 hourly response recordings or until the block has clearly worn off.

A successful response to a diagnostic injection requires documentation of positive functional changes by trained personnel and may include nurses, physician assistants, medical assistants, therapists, or non-injectionist physicians. Functional progress supersedes pain improvement.

Recommendation 109. If the diagnostic response to the initial block is considered positive, then a separate confirmatory block using a local anesthetic of different duration on a different date must be performed to confirm the level of involvement prior to rhizotomy.

RF Neurotomy Requirements.

Recommendation 110. RF neurotomy is not recommended for patients with non-facetogenic pain generators or involvement of more than 3 levels of medial branch nerves per side.

Recommendation 111. RF neurotomy is only indicated for those with proven facetogenic joint pain who have met the criteria for a positive response to 2 controlled medial branch blocks, as defined in recommendation 108 ([table 33](#)). Intra-articular facet joint steroid injections are not considered diagnostic blocks for the purposes of this criteria.

Recommendation 112. Cooled RF denervation, pulsed RF, dorsal nerve root ganglion RF ablation, and transdiscal biacuplasty are not recommended for the cervical spine.

Repeat RF Neurotomy Requirements.

Recommendation 113. Repeat RF neurotomy may be indicated if the patient experiences sustained (6 months or more), measurable, and clinically meaningful improvement in at least 3 objective functional measures (e.g., spinal ROM; reaching and lifting) and improvement in an accepted pain scale (e.g., VAS or NRS) after initial RF neurotomy ([Appendix Functional Scales](#)). Functional progress supersedes pain improvement. The potential for atrophy of the spinal musculature should be factored into clinical decision making.

Recommendation 114. If the patient's pain presents differently than the initial rhizotomy, a confirmatory medial branch block, as outlined in recommendation 108, will be necessary.

Evidence Tables.

Table 33.		
<i>Evidence Table: Cervical RF Neurotomy</i>		
Some evidence	Evidence statement	Design
	RF neurotomy relieves pain and restores function in patients whose neck pain arises from the facet joint but benefits beyond 1 year are not yet established. This was determined by a methodologically carefully done study whose application as evidence is limited by the size of the patient sample.	RCT

Section 8.a.v. Prolotherapy

Introduction. Prolotherapy, also known as sclerotherapy, consists of a series of injections of hypertonic dextrose, with or without glycerine and phenol, into the ligaments of the neck.

Contraindications / Complications / Side Effects and Adverse Events.

Side Effects and Adverse Events Related to Cervical Prolotherapy.

- Side effects and adverse events include a temporary increase in neck pain and stiffness post-injection.

Recommendations.

Core Requirements.

Recommendation 115. Prolotherapy is not recommended for neck pain.

Evidence Tables. None. The above recommendation was based on consensus.

Section 8.a.vi. Trigger Point Injection

Introduction. Trigger point injections consist of injection of local anesthetic, with or without corticosteroid, into highly localized, extremely sensitive bands of skeletal muscle fibers that produce local and referred pain when activated.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Trigger Point Injection.

- Contraindications include underlying structural issues that more reasonably account for myofascial pain or infection.

Complications of Trigger Point Injection.

- Complications include infection, pneumothorax, anaphylaxis, penetration of the viscera, neurapraxia, neuropathy, syncope, and/or local myopathy (when corticosteroid is injected).

Side Effects and Adverse Events Related to Trigger Point Injection.

- Side effects and adverse events include localized pain.
- If a patient experiences severe pain, there is a possibility of intraneural injection and the needle should be immediately repositioned.

Recommendations.

Core Requirements.

Recommendation 116. Trigger point injections should be reserved for patients who meet all of the following criteria:

- consistent, well circumscribed trigger points with a local twitch response;
- characteristic radiation of pain pattern and local autonomic reaction (e.g., persistent hyperemia following palpation); and
- trigger points are not responding to specific, noninvasive, myofascial interventions within a 6-week time frame.

Recommendation 117. Concurrent participation in a therapeutic exercise program is required while undergoing myofascial interventions. See the Active Therapies section.

Recommendation 118. Conscious sedation is not recommended for patients receiving trigger point injections.

Time Frames.

Time Frames for Trigger Point Injection			
Time to produce effect	Frequency	Optimum duration	Maximum duration
Local anesthetic - up to 30 mins No local anesthetic - up to 48 hrs	Weekly*	4 weeks	8 weeks†
*No more than 4 injections per session per week to avoid significant post-injection soreness. †Occasional patients may require 2 to 4 repetitions of trigger point injections series over a 1 to 2 year period.			

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.a.vii. Botulinum Toxin Injection

Introduction. Botulinum toxin injections are used to temporarily weaken or paralyze muscles. These injections may reduce muscle pain in conditions associated with spasticity or dystonia. See the Mild Traumatic Brain Injury MTGs for botulinum toxin injection use in the treatment of cervicogenic and migraine headaches.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute Contraindications to Botulinum Toxin Injection.

- Absolute contraindications to botulinum toxin injection include:
 - hypersensitivity to any botulinum toxin preparation or any of the components in the formulation,
 - infection at the proposed injection site, and
 - motor neuropathy or neuromuscular junction disorders.

Complications of Botulinum Toxin Injection.

- Complications include infection at the injection site.

Side Effects and Adverse Events Related to Botulinum Toxin Injection.

- Side effects and adverse events include flu-like symptoms, dry mouth, dysphagia, weakening of the paracervical musculature, and localized muscular tenderness at the site of injection.
- Neutralizing antibodies develop in at least 4% of patients treated with botulinum toxin type A, rendering it ineffective. Several antigenic types of botulinum toxin have been described. Botulinum type B appears to be effective in patients who have become resistant to the type A toxin. The immune responses to botulinum toxins type A and B are not cross-reactive, allowing type B toxin to be used when type A action is blocked by antibodies (table 34).

Recommendations.

Core Requirements.

Recommendation 119. Botulinum trigger point injections are not recommended for nonspecific neck pain or myofascial trigger points (table 35).

Recommendation 120. Botulinum toxin injection may be used for centrally mediated spasticity, dystonia, or torticollis resulting in limited ROM.

Recommendation 121. After appropriate dose titration, repeat botulinum injections are only recommended when the first injection results in at least 80% improvement in an accepted pain scale (e.g., VAS or NRS) with associated improvement in currently impaired physical function. Functional progress supersedes pain improvement.

Time Frames.

Time Frames for Botulinum Toxin Injections		
Time to produce effect	Duration between injections	Maximum duration
24 to 72 hours post-injection*	> 3 months**	Maximum 4 injections
*Peak effect by 4 to 6 weeks. **Patient should be reassessed after each injection session.		

Evidence Tables.

Table 34.		
Evidence Table: Botulinum Adverse Events		
Summary: Botulinum toxin injections commonly lead to adverse effects.		
Good evidence	Evidence statement	Design
	Cervical botulinum toxin A injections cause transient dysphagia and neck weakness. Allergic reaction to medications, dry mouth, and vocal hoarseness may also occur. Dry mouth and dysphagia occur 15% of the time after 1 injection of botulinum toxin type B.	Meta analysis
	Among those with cervical dystonia, treatment with botulinum toxin type A is more likely than placebo to lead to adverse effects such as local weakness and dysphagia.	Systematic review
	In non-naïve botulinum toxin populations, the occurrence of the specific adverse effect of "sore-throat/dry-mouth" was 36.2% higher in patients using botulinum toxin B as compared to patients using botulinum toxin A.	Systematic review

Table 35.

Evidence Table: Botulinum Injection Effectiveness

Summary:

For patients with cervical dystonia, botulinum toxin injections reduce dystonia and pain. They do not provide benefit for nonspecific cervical pain.

Table 35 continued.

Evidence Table: Botulinum Injection Effectiveness

Strong evidence	Evidence statement	Design
	Botulinum toxin A has objective and symptomatic benefits over placebo for cervical dystonia.	Meta analysis
Good evidence	Evidence statement	Design
	A single injection of botulinum toxin type B is more effective than placebo in alleviating the severity and pain of idiopathic cervical dystonia. The duration of effect of botulinum toxin type B is not certain but appears to be approximately 12 to 18 weeks.	Meta analysis
	Botulinum toxin is not different from placebo for cervical pain.	Meta analysis

Section 8.b. Surgical Interventions

Introduction. Operative treatments are considered when the natural history for surgically treated lesions is better than that of non-operatively treated lesions. Most individuals with neck pain will not need a surgical intervention. See the [Overview of Care](#) section

Contraindications / Complications / Side Effects and Adverse Events.

- See specific surgical procedure section.

Recommendations.

Core requirements.

Recommendation 122. Expedited surgical referral is required for patients with myelopathy, acute fracture, or dislocation or for patients with functional loss due to stenosis and instability and/or progressive neurologic deficits.

Recommendation 123. Surgical referral for patients with cervical radiculopathy is recommended when confirmatory imaging studies specifically correlate with clinical findings, demonstrating nerve root compression or spinal cord compromise and any of the following are present:

- acute, incapacitating pain with progressive neurologic deficits;
- persistent or recurrent arm pain with functional limitations unresponsive to 6 weeks of treatment;
- progressive neurologic deficits; or
- static neurologic deficit with significant radicular pain.

Recommendation 124. Surgical referral for patients with persistent non-radicular cervical pain is recommended when frequent symptom recurrence results in functional limitations even if non-operative treatment provides temporary relief and all of the following are present:

- non-operative treatment fails to result in consistent functional improvement;
- diagnostic elimination of confounding psychological or physical conditions that may respond to non-surgical techniques but may be refractory to surgical intervention; and
- symptomatic improvement has plateaued and residual functional disability is unacceptable at 6 months.

Recommendation 125. Continuation of exercise and/or active therapy while awaiting a surgical evaluation or surgical procedure is recommended to maintain and optimize physical conditioning.

Recommendation 126. Informed consent is required before all invasive procedures. See the Essential First Line Treatment section.

Recommendation 127. Participation in a postoperative active therapy program that includes core stabilization, strengthening, and endurance is strongly recommended. See the Active Therapies section.

Recommendation 128. Diabetes control and related routine lab work, including hemoglobin A1c, are recommended prior to surgery.

Recommendation 129. A multi-modal approach to postoperative pain management is recommended to avoid overuse and misuse of opioid medications. This approach may include any of the following:

- cognitive behavioral therapy (CBT) (see the Behavioral and Psychological Interventions section);
- pain neuroscience education (see the Active Therapies section);
- use of non-opioid medication strategies as adjunct to anesthesia and pre- or postoperative pain management (see the Medications section);
- use of peripheral regional analgesia; and/or
- use of passive therapy for postoperative pain (see the Passive Therapies section).

Recommendation 130. Intraoperative neuromonitoring is permitted to evaluate spinal cord integrity and screw placement during the operative procedure.

Recommendation 131. Percutaneous RF disc decompression and intradiscal electrothermal therapy are not recommended.

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.b.i Discectomy

Introduction. Discectomy is a surgical treatment for ruptured or herniated discs of the cervical spine. There are several procedures in use to decompress extruded disc material around nerve roots, including several minimally invasive techniques. Cervical discectomy is typically paired with fusion. Please see the Spinal Fusion section for additional information. This includes:

- microscopic discectomy,
- percutaneous discectomy,
- microdiscectomy, and
- laser discectomy.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Cervical Discectomy.

- Complications of discectomy include nerve damage, spinal fluid leakage, esophageal injury, infection, vascular injury, and hemorrhage.
- Complications of percutaneous discectomy include nerve or vessel injury, infection, hematoma, and incomplete nerve root dissection.

Recommendations.

Core Requirements.

Recommendation 132. A psychological screen with a follow-up psychological evaluation, if indicated, is required prior to proceeding with discectomy. If the surgery is being performed in an emergent or urgent situation, the screen is not necessary.

Recommendation 133. For a patient to be a candidate for an isolated cervical discectomy, the provider must document all of the following:

- subjective report of severe radicular pain;
- positive neural compression test (e.g., dural tension, Spurling's sign, traction/distraction, upper limb tension test) and/or reflex, motor, or sensory changes on examination consistent with imaging findings; and
- objective findings of neural compression on imaging studies that specifically correlate with the subjective report and examination findings.

Recommendation 134. ESIs can be trialed prior to surgery if the patient has a preference to avoid surgery, is unable to participate in therapy after the first 2 weeks post-injury, and meets the criteria for ESI outlined in [recommendation 95](#).

Recommendation 135. The decision to proceed with an open or minimally invasive technique for cervical discectomy is left to the treating surgeon's discretion, experience, and training in these techniques ([table 36](#)).

Acute Phase Requirements (Within 6 Weeks of Initial Care).

Recommendation 136. Early referral within 6 weeks for isolated cervical discectomy is permitted when the provider documents all criteria in recommendations [132](#) and [133](#) and at least 1 of the following:

- radicular pain-related functional impairment prevents participation in active therapy and ADLs;
- presence of progressive functional neurological deficits; or
- an opioid treatment is being considered.

Subacute and Chronic Phase Requirements (After 6 Weeks of Care).

Recommendation 137. Referral for isolated cervical discectomy is permitted for patients who meet the criteria in [recommendation 133](#) and both of the following:

- functional impairment despite 6 weeks of optimal medical management and participation in active therapy; and
- confounding psychosocial risk factors have been screened for and clinically addressed (see the Behavioral and Psychological Interventions section).

Recommendation 138. Laser discectomy and percutaneous discectomy are not recommended.

Evidence Table.

Table 36.		
<i>Evidence Table: Effectiveness of Cervical Discectomy</i>		
Some evidence	Evidence statement	Design
	Minimally invasive discectomy has not been shown to improve function, extremity pain, or axial pain compared to open discectomy in the setting of cervical and lumbar primary discectomy.	Systematic review

Section 8.b.ii. Decompression for Central or Lateral Spinal Stenosis

Introduction. Multiple procedures described in this section offer surgical access to decompress neural elements by partial or total removal of various parts of the spinal elements. These procedures include:

- laminotomy,
- laminectomy,
- foraminotomy,
- facetectomy,
- corpectomy,
- discectomy, and
- laminoplasty.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Decompressive Surgery.

- Unaddressed depression or cardiovascular comorbidity are relative contraindications.
- Documented radiographic instability is a contraindication for decompression alone.
- Scoliosis is a relative contraindication to decompression.

Complications of Decompressive Surgery.

- Complications include graft dislodgement, nerve injury, post-surgical instability, cerebrospinal fluid leakage, hematoma, hemorrhage, infection, spinal cord injury causing paralysis, perineural fibrosis, ROM loss, kyphosis, and incomplete decompression.

Recommendations.

Core Requirements.

Recommendation 139. A psychological screen with a follow-up psychological evaluation, if indicated, is required prior to proceeding with decompression. If the surgery is being performed in an emergent or urgent situation, the screen is not necessary.

Recommendation 140. Surgical indications include all of the following:

- radicular symptoms or symptoms of neurogenic claudication, often with clinical evidence of radiculopathy that correlates with the patient's pain and findings;
- evidence of nerve root or spinal compression on imaging studies; and
- failure of non-surgical care, including a trial of 6 weeks of active therapy.

Recommendation 141. Cervical laminoplasty or laminectomy alone is recommended for patients with spinal stenosis or spondylotic myelopathy primarily related to posterior spinal compressive pathology in the absence of kyphosis.

Recommendation 142. Corpectomy or hemicorpectomy with fusion is recommended for patients with 1- or 2- level spinal stenosis, spondylolisthesis, or severe kyphosis and associated spinal cord compression.

Evidence Tables. None. The above recommendations were based on consensus.

Section 8.b.iii. Spinal Fusion

Introduction. Spinal fusion is a procedure that unites 2 or more vertebral bodies together to restrict motion and removes a degenerative disc to relieve symptoms of coexistent nerve root compression. The procedure often involves the use of bone grafts, and sometimes instrumentation, to produce a rigid connection between 2 or more adjacent vertebrae.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Cervical Fusion.

Morbid obesity is a relative contraindication to fusion ([table 37](#)).

Complications of Cervical Fusion.

Complications include instrumentation failure, pseudoarthrosis, superficial or deep wound infection, incomplete decompression, in-hospital mortality, bone graft donor site pain graft extrusion, cerebrospinal fluid leak, laryngeal nerve damage, iatrogenic kyphosis, dysphonia, dysphagia, esophageal perforation, airway obstruction, and acceleration of adjacent segment disease ([table 37](#)).

Recommendations.

Core Requirements.

Recommendation 143. Cervical fusion is not recommended in the first 6 months of symptoms, except for acute fracture, dislocation, or for some patients with functional loss due to stenosis and instability, myelopathy, and/or progressive neurologic deficits.

Recommendation 144. Psychological evaluation is required to assess suitability for a cervical fusion. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes; or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section ([table 38](#)).

Recommendation 145. Spinal fusion is reserved for patients who meet either of the following sets of criteria:

either

- cervical radiculopathy resulting in incapacitating pain; and

- imaging studies (e.g., MRI) consistent with clinical findings, demonstrating nerve root or spinal cord compromise; and
- one of the following:
 - progressive functional neurological deficit; or
 - persistent motor deficit; or
 - persistent or recurrent arm pain with functional limitations, unresponsive to conservative treatment after 6 weeks; or
 - static neurological deficit associated with significant radicular pain.

or

- persistent non-radicular cervical pain unrelieved by non-operative treatment; and
- all pain generators are adequately defined and treated; and
- all physical medicine and manual therapy interventions are completed; and
- imaging studies demonstrate spinal stenosis with instability or disc pathology, requiring decompression; and
- psychological evaluation, with confounding issues addressed as discussed in recommendation 144; and
- one of the following:
 - improvement of symptoms has plateaued and the residual symptoms of pain and signs of functional disability are unacceptable at the end of 6 months of active treatment; or
 - frequent symptom recurrence causes serious functional limitations even if a non-operative active treatment program provides satisfactory symptom relief and functional restoration at each occurrence (tables 39, 40).

Recommendation 146. In the absence of radiculopathy, fusion is limited to 2 spinal levels.

Recommendation 147. Discography is not recommended for use in evaluating cervical spine conditions.

Recommendation 148. Tobacco cessation for at least 6 weeks prior to fusion is required, unless the surgeon documents reasoning that the benefits of proceeding with surgery outweigh the risks.

Recommendation 149. Cervical application of recombinant human bone morphogenetic protein (rhBMP) is not recommended (table 37).

Recommendation 150. The choice of bone graft donor site, grafting substance, operative technique, and surgical approach are at the discretion of the treating surgeon (tables 41, 42).

Recommendation 151. Strategic perioperative medication use is recommended for pain management to minimize opioid use and optimize surgical outcome.

Recommendation 152. Postoperative participation in an active therapy program that includes core stabilization, strengthening, and endurance is recommended. See the Active Therapies section.

Recommendation 153. Cervical bracing is permitted following fusion.

Evidence Tables.

Table 37.

Evidence Table: Complications Related to Cervical Fusion

Good evidence	Evidence statement	Design
	Recombinant human bone morphogenetic protein-2 (rhBMP-2) increases the risks of dysphagia, dysphonia, and other postoperative complications in the setting of anterior cervical fusion, but there is uncertainty regarding the risks of cancer associated with the use of bone morphogenetic protein (BMP) in spine fusion of any type and uncertainty about the comparative risks of retrograde ejaculation sometimes attributed to BMP.	Meta analysis
Some evidence	Evidence statement	Design
	Morbid obesity increases hospital length of stay, mortality, and postoperative complications after spinal fusion surgery, with concomitant increases in hospital costs.	Cross sectional study

Table 38.

Evidence Table: Factors Influencing Surgical Outcomes

Some evidence	Evidence statement	Design
	Depression at the time of surgery and in the early recovery period is associated with poorer functional recovery at 2 years, even though it does not appear to be associated with worse pain at 2 years.	Cohort study

Table 39.

Evidence Table: Anterior Cervical Discectomy and Fusion (ACDF)

Strong evidence	Evidence statement	Design
	Outpatient anterior cervical discectomy and fusion (ACDF) and inpatient ACDF both have low postoperative complication rates, and there is no difference in risks of commonly occurring complications for the procedure between the 2 settings, but there is insufficient evidence to compare the safety of the 2 settings for rare but potentially catastrophic events.	Meta analysis
Good evidence	Evidence statement	Design
	In the setting of symptomatic cervical radiculopathy from herniated discs and/or osteophytes at 1 or 2 levels, it is reasonable to initiate a program of structured physical therapy before any surgical decisions are made, since many patients will experience significant improvement in pain and function. However, a program of ACDF plus structured physical therapy is likely to be superior to physical therapy alone, and the improvements appear to continue for at least 5 to 8 years.	RCT

Table 40.

Evidence Table: ACDF Versus Total Disc Arthroplasty

Summary:

Both cervical disc arthroplasty (CDA) and ACDF are safe and effective interventions for patients with cervical degenerative disc disease. CDA leads to greater improvements in pain and disability and fewer reoperations than ACDF.

Strong evidence	Evidence statement	Design
	For patients with cervical degenerative disc disease at 1 or 2 levels, both CDA and ACDF are safe and effective interventions leading to symptomatic relief and functional improvement.	Meta analysis
	In patients with single level radiculopathy or myelopathy, cervical artificial disc produces 2-year success rates at least equal to those of ACDF with allograft interbody fusion and an anterior plate.	Meta analysis
	For patients with single-level cervical degenerative disc disease, there is strong evidence that both CDA and ACDF are safe and effective interventions leading to symptomatic relief and functional improvement.	Systematic review
Good evidence	Evidence statement	Design
	Overall success, ascertained by improvement in neck disability scores combined with favorable neurological outcome and no repeat surgery at the operated level, is better with CDA than with ACDF.	Meta analysis

Table 40 continued.

Evidence Table: ACDF Versus Total Disc Arthroplasty

Good evidence cont.	Evidence statement	Design
	In patients with stable 1 or 2 level cervical disc disease who have minimal facet joint degeneration and bone mineral density greater than a T-score of -1.5, and who do not have axial neck pain as the sole presenting symptom, treatment with either total disc replacement (TDR) or ACDF are likely to alleviate pain and disability for at least 7 years, and that the TDR procedure is likely to confer greater benefit than fusion. Reoperation rates may be lower with disc replacement but this inference must be made with considerable caution.	RCT
	CDA leads to slightly better outcomes than ACDF for improvements in neck and arm pain and cervical spine function.	Systematic review
	CDA is associated with fewer repeat operations at adjacent spine levels than ACDF for the first several years after the surgery is performed.	Systematic review
	Evidence statement	Design
	In the setting of radiographically stable cervical degenerative disc disease from C3 to C7 at 2 adjacent levels, CDA is at least as safe and effective as fusion with an anterior plate and a ring allograft, and is likely to require fewer reoperations in the first 2 years after surgery.	RCT

Some evidence	Evidence statement	Design
	In patients with single-level cervical radiculopathy, without signs of myelopathy and with preservation of at least 3mm of disc height, equally good improvements in neck disability are expected after 2 years with any of 3 operations: anterior cervical discectomy alone, anterior cervical discectomy with fusion and a polyetheretherketone (PEEK) cage, or anterior CDA. Evidence is lacking to suggest that the development of adjacent segment degeneration is significantly different after 2 years with 1 or another of these 3 surgical options.	RCT
	Cervical TDR requires fewer revision operations than ACDF after the first 2 years of treatment, and TDR slightly decreases neck pain at 5 years compared to ACDF. Half of the reoperations in the ACDF group were at adjacent levels.	RCT

Table 41.

Evidence Table: Cervical Fusion Graft Site

Good evidence	Evidence statement	Design
	In the setting of ACDF, P-15 (trademark name i-Factor), a 15 amino acid peptide, is noninferior to local bone autograft for achieving radiographic fusion, improvement in neck disability, and neurologic success, and has a safety profile similar to that of autograft.	RCT

Table 42.

Evidence Table: Cervical Fusion - Other Technical Issues

Good evidence	Evidence statement	Design
	In the setting of spine fusion surgery, the use of antifibrinolytics such as tranexamic acid reduces intraoperative blood loss and total blood loss, leading to lower transfusion requirements, while not being accompanied with significant adverse effects compared to no antifibrinolytic use.	Systematic review
Some evidence	Evidence statement	Design
	In cervical fusion for degenerative disease, iliac crest autograft provides greater fusion rates, but cages are a valid alternative as cages result in fewer complications from surgery.	Meta analysis
	In the setting of ACDF, a zero-profile implant is less likely to lead to postoperative dysphagia than a conventional cervical plate for up to 1 year after the operation.	Meta analysis
	In the setting of ACDF, a PEEK cage filled with bone graft substitute in the form of calcium sulfate and demineralized bone matrix achieves similar rates of fusion and clinical success as a PEEK cage filled with iliac crest autograft, while eliminating morbidity related to the donor site.	RCT

Section 8.b.iv. Cervical Total Disc Replacement (TDR)

Introduction. Cervical total disc replacement (TDR) is a surgical procedure where a degenerated disc is replaced with a prosthetic device after decompression of the neural elements is completed. The endplates are positioned under intraoperative fluoroscopic guidance for optimal placement in the sagittal and frontal planes. The prosthetic device physiologically distributes the mechanical load of the vertebrae and maintains ROM.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Cervical TDR.

- Absolute and relative contraindications include:
 - significant spinal deformity and scoliosis;
 - symptomatic facet joint arthrosis;
 - osteopenia, osteoporosis, or any metabolic bone disease;
 - spinal instability at the pathologic level;
 - spinal instability at an adjacent level that is not appropriate for fusion;
 - deficient posterior elements;
 - infection;
 - previous compression or burst fracture;
 - spondylolisthesis > 3 mm;
 - spondylolysis;
 - chronic steroid or other medication use that interferes with bone or soft tissue healing;
 - allergy to device materials;
 - morbid obesity (e.g., body mass index > 40 kg/m² or over 100 pounds overweight);
 - active malignancy; or
 - generalized chronic pain.

Complications of Cervical TDR.

- Complications include nerve and vascular injury, dural tears, malpositioning/suboptimal positioning of prosthesis, implant failure, complex regional pain syndrome, and need for reoperation.

Recommendations.

Core Requirements.

Recommendation 154. Psychological evaluation is required to assess suitability for cervical disc replacement. Documentation should include the following items with associated treatment recommendations:

- psychological factors that might influence elective surgical treatment outcomes, or
- psychological factors that might complicate surgical recovery.

Confounding depression or anxiety must be addressed prior to proceeding with surgery. Presurgical psychological evaluation should not be done by a psychologist employed by the physician performing the procedure. See the Behavioral and Psychological Interventions section.

Recommendation 155. Cervical TDR is reserved for patients who meet either of the following sets of criteria:

either

- symptomatic 1- or 2- level disc disease established by objective testing (e.g., CT scan or MRI) with established radiculopathy or myelopathy unrelieved after 6 weeks of active nonsurgical treatment; and
- radiculopathy or myelopathy documented by electromyography or MRI with correlated objective findings;

or

- symptoms unrelieved after 6 months of active nonsurgical treatment; and
- all pain generators are adequately defined and treated; and
- all physical medicine and manual therapy interventions are completed; and
- imaging studies demonstrate disc pathology requiring decompression; and
- psychosocial evaluation as outlined in recommendation 154, with confounding issues addressed (table 43).

Recommendation 156. If there are signs suggestive of facet-mediated pain, the pain must be evaluated with medial branch blocks before proceeding with disc replacement.

Recommendation 157. A trial of multi-disciplinary therapy must occur prior to proceeding with surgery.

Recommendation 158. Surgeons performing total disc arthroplasty must have attended training courses or undergone disc replacement training in fellowship. It is highly recommended that the surgeon is mentored and proctored.

Evidence Tables.

Table 43.

Evidence Table: Total Disc Arthroplasty Versus Spinal Fusion

Summary:

Both CDA and ACDF are safe and effective interventions for patients with cervical degenerative disc disease. CDA leads to greater improvements in pain and disability, and fewer reoperations, than ACDF.

Strong evidence	Evidence statement	Design
	For patients with cervical degenerative disc disease at 1 or 2 levels, both CDA and ACDF are safe and effective interventions leading to symptomatic relief and functional improvement.	Meta analysis

Table 43 continued.

Evidence Table: Total Disc Arthroplasty Versus Spinal Fusion

Strong evidence cont.	Evidence statement	Design
	In patients with single level radiculopathy or myelopathy, cervical artificial disc produces 2-year success rates at least equal to those of ACDF with allograft interbody fusion and an anterior plate.	Meta analysis
	For patients with single-level cervical degenerative disc disease, there is strong evidence that both CDA and ACDF are safe and effective interventions leading to symptomatic relief and functional improvement.	Systematic review

Good evidence	Evidence statement	Design
	CDA is associated with fewer repeat operations at adjacent spine levels than ACDF for the first several years after the surgery is performed.	Systematic review
	CDA leads to slightly better outcomes than ACDF for improvements in neck and arm pain and cervical spine function.	Systematic review
	In patients with stable 1 or 2 level cervical disc disease who have minimal facet joint degeneration and bone mineral density greater than a T-score of -1.5, and who do not have axial neck pain as the sole presenting symptom, treatment with either TDR or ACDF are likely to alleviate pain and disability for at least 7 years, and that the TDR procedure is likely to confer greater benefit than fusion. Reoperation rates may be lower with disc replacement but this inference must be made with considerable caution.	RCT
	Overall success, ascertained by improvement in neck disability scores combined with favorable neurological outcome and no repeat surgery at the operated level, is better with CDA than with ACDF.	Meta analysis
	In the setting of radiographically stable cervical degenerative disc disease from C3 to C7 at 2 adjacent levels, CDA is at least as safe and effective as fusion with an anterior plate and a ring allograft, and is likely to require fewer reoperations in the first 2 years after surgery.	RCT
Some evidence	Evidence statement	Design
	Cervical TDR requires fewer revision operations than ACDF after the first 2 years of treatment, and TDR slightly decreases neck pain at 5 years compared to ACDF. Half of the reoperations in the ACDF group were at adjacent levels.	RCT
	In patients with single-level cervical radiculopathy, without signs of myelopathy and with preservation of at least 3mm of disc height, equally good improvements in neck disability are expected after 2 years with any of 3 operations: anterior cervical discectomy alone, anterior cervical discectomy with fusion and a PEEK cage, or anterior CDA. Evidence is lacking to suggest that the development of adjacent segment degeneration is significantly different after 2 years with 1 or another of these 3 surgical options.	RCT

Section 8.b.v. Spinal Cord Stimulators

Introduction. Spinal cord stimulation is the delivery of low-voltage electrical stimulation to the spinal cord or peripheral nerves to inhibit or block the sensation of pain. The system uses implanted electrical leads and a battery powered implanted pulse generator. See the Chronic Pain Disorder MTGs.

Recommendations.

Core Requirements.

Recommendation 159. Neurostimulation is reserved for patients who meet the criteria outlined in the Chronic Pain Disorder MTGs.

Evidence Tables. See the Chronic Pain Disorder MTGs for evidence.

Section 8.b.vi. Epiduroscopy with Epidural Lysis of Adhesions

Introduction. Epiduroscopy and epidural lysis of adhesions involves the introduction of a fiberoptic endoscope into the epidural space. A saline irrigation is performed with or without epiduroscopy to distend the epidural space and result in lysis of adhesions.

Contraindications / Complications / Side Effects and Adverse Events.

Complications of Epiduroscopy.:

- Saline irrigation on epiduroscopy and lysis of adhesions is associated with risks of elevated pressures that may impede blood flow and venous return, possibly causing ischemia of the cauda equina and retinal hemorrhage.
- Other epiduroscopy complications associated with instrumented lysis include catheter shearing, need for catheter surgical removal, infection (including meningitis), hematoma, and possible severe hemodynamic instability during application.

Recommendations.

Core Requirements.

Recommendation 160. Epiduroscopy and mechanical lysis of adhesions and epiduroscopy-directed steroid injections are not recommended.

Evidence Tables. See the [Chronic Pain Disorder MTGs](#) for evidence.

Section 9. Medications

Introduction. Medications are used in the treatment of neck injuries to control acute pain, chronic pain, and inflammation. Use of medications will vary widely due to the spectrum of injuries. If medications are being considered for long-term chronic pain management, refer to the Chronic Pain Disorder Medical Treatment Guidelines (MTGs) medication section.

Contraindications / Complications / Side Effects and Adverse Events.

- See the specific medication section. The medication lists within these sections do not provide complete information on side effects, potential complications, drug interactions, or drug monitoring. For more complete information, refer to the Chronic Pain Disorder MTGs medication section or a medication reference text.

Recommendations.

Core Requirements.

Recommendation 161. Medication reconciliation is required at the initial visit and periodically during treatment to avoid medication errors and to discuss side effects, drug interactions, and expected functional goals. Reconciliation includes the following elements:

- current medication name, dosage, frequency, and route;
- patient understanding of indication;
- potential interaction of prescription and over the counter medications;
- drug allergies;
- comorbid medical issues;
- history of substance abuse; and
- checking the Colorado Prescription Drug Monitoring Program (PDMP).

The medications documented as a part of the reconciliation will reflect those that the provider deems directly relevant to the claim-related condition.

Recommendation 162. A therapeutic trial of medications is recommended to evaluate the effect on functional status. The length of a medication trial will depend on the individual medication, and the patient should be informed on the time to expected benefit. If no functional benefit is observed at that time, the medication should be discontinued.

Recommendation 163. Medications should be initiated at the lowest dose expected to result in functional improvement and then titrated based on clinical response.

Evidence Tables. None. The above recommendations were based on consensus.

Section 9.a. Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) and Acetaminophen

Introduction. Non-steroidal anti-inflammatory drugs (NSAIDs) are medications for neck pain and inflammation. Acetaminophen is an analgesic commonly used to treat mild pain.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Acetaminophen and NSAIDs.

- Acetaminophen

- Concomitant use of other drug products containing acetaminophen exceeding the maximum recommended daily dose of acetaminophen.
- Allergy to acetaminophen or any of the inactive ingredients.
- Use with caution in patients with alcoholic liver disease, patients with hepatic impairment or active liver disease, and patients with known glucose-6-phosphate-dehydrogenase (G6PD) deficiency.
- NSAIDs ([table 44](#))
 - Comorbid conditions that result in an increased risk for gastrointestinal (GI), renal, or cardiovascular adverse reactions.
 - Patients at risk for GI bleeding, including patients who use alcohol, smoke, are older than 65 years old, take corticosteroids or anti-coagulants, and patients on longer durations of therapy.
 - Celecoxib is contraindicated in sulfonamide allergic patients.
 - Contraindicated in patients with aspirin-exacerbated respiratory disease due to cross-reactivity risk and potential for a severe asthma attack.
- Topical salicylate and non-salicylate medications may result in alterations in bleeding time; they should be used with caution in patients on warfarin therapy.

Side Effects and Adverse Events Related to Specific Classes of Acetaminophen and NSAIDs.

- Acetaminophen may result in hepatotoxicity, and chronic use may result in chronic kidney disease, hypersensitivity or anaphylactic reactions, serious and potentially fatal skin reactions, hypertension, chronic daily headaches, and peptic ulcer disease.
- NSAIDs may result in abnormal or worsening renal function, including renal failure; abnormal liver function; GI bleeding, particularly in patients at higher risk for a bleed; increased risk of cardiovascular events; gastric or duodenal perforation and ulceration; anaphylactoid reaction; platelet function abnormalities; and fluid retention and edema. Postoperative NSAIDs may increase the risk of nonunion at higher doses.
- Topical agents may result in localized skin reactions.

Recommendations.

Core Requirements.

Recommendation 164. Acetaminophen or NSAIDs are recommended for initial analgesic treatment of uncomplicated neck pain ([tables 45, 46](#)).

Recommendation 165. Co-prescription of a proton pump inhibitor, histamine H2-receptor antagonists (H2-blockers), or prostaglandin analog with NSAIDs is recommended to reduce risk of duodenal or gastric ulceration in patients with concurrent antiplatelet or corticosteroid therapy.

Recommendation 166. Cyclooxygenase-2 (COX-2) inhibitors are not recommended as a first-line agent for short-term use in low-risk patients, but they can be used for patients who do not tolerate traditional NSAIDs.

Recommendation 167. Topical NSAIDs are recommended when oral NSAID use is contraindicated due to systemic side effects. It must be started with the lowest dose anticipated to achieve the desired clinical effect and then titrated until functional improvement is noted ([table 47](#)).

Recommendation 168. Perioperative use of acetaminophen and/or NSAIDs, either alone or in combination with other medications, to optimize analgesia is recommended.

Recommendation 169. See the [Chronic Pain Disorder MTGs medication section](#) if acetaminophen or NSAIDs are being considered for long-term use.

Time Frames.

Time Frames for Medications		
	Optimum duration	Maximum duration
Acetaminophen	up to 10 days	Extended use on a case-by-case basis*
NSAIDs	7 days	1 year*
*Use of this substance long-term (for 3 days per week or greater) may be associated with rebound pain upon cessation.		

Evidence Tables.

Table 44.

Evidence Table: Adverse Events and NSAIDs

Good evidence	Evidence statement	Design
	Naproxen has a more favorable cardiovascular profile than other NSAIDs when used over a long period for chronic pain.	Meta analysis
	Celecoxib in a dose of 200 mg per day, administered over a long period, does not have a worse cardiovascular risk profile than naproxen at a dose of up to 1000 mg per day or ibuprofen at a dose of up to 2400 mg per day.	RCT
	Celecoxib has a more favorable safety profile than ibuprofen or naproxen with respect to serious GI adverse events, and has a more favorable safety profile than ibuprofen with respect to renal adverse events.	RCT
Some evidence	Evidence statement	Design
	Fewer adverse events occurred in NSAIDs treatment arms as compared to placebo in patients experiencing chronic low back pain.	Systematic review
	Topical NSAIDs are associated with fewer systemic adverse events than oral NSAIDs.	Meta analysis

Table 45.

Evidence Table: Acetaminophen

Strong evidence	Evidence statement	Design
	In the setting of acute nonspecific low back pain, acetaminophen at a dose of up to 4 grams per day is no more effective than placebo for pain relief and for improvement of disability.	Meta analysis

Table 46.

Evidence Table: NSAIDs

Summary:

NSAIDs may reduce acute low back pain and short term disability when compared to placebo, but the clinical benefit is small. Postoperative NSAIDs may increase the risk of nonunion at higher doses. While the literature in this table discusses low back pain, there is presumed physiologic similarity for neck pain.

Good evidence	Evidence statement	Design
	For persistent low-back pain, oral NSAIDs are more effective than placebo or acetaminophen.	Systematic review

Table 46 continued.

Evidence Table: NSAIDs

Good evidence cont.	In the setting of lumbar spinal fusion, NSAIDs at an equivalence of no more than 120 mg of ketorolac for 14 days postoperatively do not appear to increase the risk of nonunion of the operated spinal segment.	Systematic review
	In the setting of acute low back pain lasting less than 12 weeks, NSAIDs lead to a small reduction in short term pain intensity, approximately 10 points on a 100 point scale, compared to placebo.	Meta analysis
Some evidence	Evidence statement	Design
	NSAIDs are not more beneficial than placebo when evaluating pain-intensity and disability outcomes; the change in these scores does not reach a minimal clinical important difference in patients experiencing chronic low back pain.	Systematic review
	In the setting of lumbar fusion without bone morphogenetic protein (BMP), postoperative NSAIDs with an equivalent dose of more than 300 mg of diclofenac are associated with an increased risk of nonunion of the operated spinal segment. There appears to be a dose-dependent effect of postoperative NSAIDs on the risk of nonunion.	Systematic review

Table 47.

Evidence Table: Topical NSAIDs for Analgesia

Strong evidence	Evidence statement	Design
	Topical NSAIDs are more effective than placebo vehicles, such as gels or creams, in the setting of acute musculoskeletal injuries.	Meta analysis

Section 9.b. Muscle Relaxants

Introduction. Muscle relaxants are a heterogeneous class of medications with varying mechanisms of action used to treat muscle spasm associated with neck injury.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Muscle Relaxants.

- Cyclobenzaprine should not be used when a patient has a history of cardiac dysrhythmia or when there is concurrent use of monoamine oxidase inhibitors.
- Metaxalone should not be used when there is a history of significantly impaired renal or hepatic disease, pregnancy, and predisposition to drug induced hemolytic anemia.
- Methocarbamol should not be used when the patient has a hypersensitivity to the medication or there is possible renal compromise.
- Tizanidine should not be used when there is concurrent use of ciprofloxacin or fluvoxamine or hepatic disease.

Side Effects and Adverse Events Related to Muscle Relaxants.

- Cyclobenzaprine may cause sedation, anticholinergic reactions, and blurred vision. Patients should also be monitored for suicidal ideation and drug abuse.
- Metaxalone may cause sedation and hematologic abnormalities.
- Methocarbamol may cause decreased cognition, lightheadedness, and GI side effects.
- Tizanidine may cause hypotension, sedation, hepatotoxicity, hallucinations, psychosis, dry mouth.

Recommendations.

Core Requirements.

Recommendation 170. Muscle relaxants may be used as an adjunct to rest and active therapy for relief of muscle spasm associated with acute, painful, musculoskeletal conditions (table 48).

Recommendation 171. Tizanidine (alpha-2 adrenergic agonist) is only approved by the Food and Drug Administration (FDA) for the treatment of true centrally mediated spasticity associated with musculoskeletal disorders. Use for musculoskeletal conditions, such as neck or back pain, would be “off label.”

Recommendation 172. Benzodiazepines are not recommended for use in combination with opioids due to the elevated risk of death due to respiratory depression.

Recommendation 173. See the Chronic Pain Disorder MTGs medication section if muscle relaxants are being considered for long-term use.

Time Frames.

Time Frames for Muscle Relaxants	
Optimum duration	Maximum duration
1 week	2 weeks [†]
[†] or longer, if only used at night.	

Evidence Tables.

Table 48.		
<i>Evidence Table: Muscle Relaxants</i>		
Summary: For patients with spine pain, muscle relaxants provide greater short-term pain relief than placebo. However, they do not provide additional benefit over NSAIDs alone, and they increase adverse events. While the literature in this table discusses low back pain, there is presumed physiologic similarity for neck pain.		
Strong evidence	Evidence statement	Design
	Non-benzodiazepine muscle relaxants are more effective than placebo for providing short-term pain relief in acute low back pain. They should be used with caution because of central nervous system side effects.	Meta analysis
Good evidence	Evidence statement	Design
	In patients with acute nontraumatic musculoskeletal low back pain who have optimized their use of NSAIDs such as naproxen, the addition of cyclobenzaprine 5mg to be taken as needed adds no significant functional benefit in the short term, but it does increase the frequency of adverse events such as dizziness, drowsiness, and GI side effects.	RCT
	In the setting of acute nonradicular, nontraumatic musculoskeletal low back pain of 2 weeks duration or less, the addition of diazepam to naproxen adds no additional pain relief or reduction of functional disability beyond that of placebo.	RCT

Section 9.c. Oral Steroids

Introduction. Oral steroids are used for pain and inflammation.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Oral Steroids.

- Absolute and relative contraindications include poorly controlled diabetes mellitus or hypertension; heart failure with peripheral edema; cataract or glaucoma; peptic ulcer disease; presence of injection; and low bone mineral density or osteoporosis.

Complications of Oral Steroids.

- Complications include gastritis, ulcer formation, and GI bleeding.

Side Effects and Adverse Events Related to Oral Steroids.

- Side effects and adverse events include fluid retention, mood disturbance, and hyperglycemia.

Recommendations.

Core Requirements.

Recommendation 174. Oral steroids are generally not recommended for the treatment of acute neck pain with or without radiculopathy (table 49).

Evidence Tables.

Table 49.		
<i>Evidence Table: Oral Steroids for Radicular Pain</i>		
Summary: For patients with radicular pain, oral steroids do not provide clinically important improvements in function or pain, nor reductions in the rate of surgery. They also increase the risk for serious adverse effects. While the literature in this table discusses radicular low back pain, there is presumed physiologic similarity for radicular neck pain.		
Good evidence	Evidence statement	Design
	In patients with an acutely herniated lumbar disc with radicular symptoms, a 15-day course of tapering oral prednisone may produce a small improvement in spine function compared to placebo, but this improvement is of very uncertain clinical importance. Also, there is no evidence that oral prednisone reduces the rate of back surgery in the following year. The benefits of a small functional improvement should be weighed against the frequent occurrence of steroid-related adverse effects such as insomnia, nervousness, and increased appetite. There is no evidence that oral prednisone reduces sciatica pain compared to placebo.	RCT
Some evidence	Evidence statement	Design
	In patients who have not taken oral steroids in the past year, short-term use of an oral steroid can increase the risk of fracture, sepsis, and venous thromboembolism in the subsequent 5 to 90 days, with the greatest increased risk occurring in the first 5-30 days after the prescription is filled. (Waljee et al., 2017)	Cohort study
Some evidence	Evidence statement	Design
	Among patients presenting to the emergency department with acute onset of musculoskeletal or radicular low back pain or acute exacerbation of chronic low back pain, a 5-day course of 50 mg oral prednisone results in similar pain reduction at 5 days post-discharge when compared to a placebo given over 5 days.	RCT
	A short course of oral corticosteroids, lasting 14 days or fewer, is associated with an increased incidence of GI bleeding, sepsis, and heart failure in the period following their administration.	Cohort study

Table 50.

Evidence Table: Other Medications

Strong evidence	Evidence statement	Design
	Bisphosphonates taken alone or in combination with calcium and/or vitamin D are more beneficial than placebo taken alone or with calcium and/or vitamin D for the prevention and treatment of spinal glucocorticoid induced osteoporosis, with data extending to 24 months of use.	Systematic review

Section 9.d. Gabapentinoids

Introduction. Gabapentinoids are anticonvulsant and nerve pain medications that may be used to treat neuropathic symptoms. They include gabapentin and pregabalin.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Gabapentinoids.

- Renal insufficiency is a relative contraindication to gabapentin use, but dosage can be adjusted to accommodate use in the setting of renal dysfunction.

Side Effects and Adverse Events Related to Gabapentinoids.

- Gabapentinoids may cause dizziness, sedation, and respiratory depression in older patients who receive gabapentin along with other analgesics or sedatives.
- Coadministration of opioids and pregabalin or gabapentin may increase the risk of opioid related mortality.
- Gabapentinoids may be associated with an increased risk of mental health disturbance (e.g., depression, suicide), unintentional overdose, and motor vehicle accidents.

Recommendations.

Core Requirements.

Recommendation 175. A trial of oral gabapentin, with a goal of using the lowest dose expected to result in functional improvement, can be used for patients with radicular cervical pain (table 51).

Recommendation 176. If a gabapentinoid is being considered for long-term, chronic pain management, refer to the Chronic Pain Disorder MTGs medication section.

Evidence Table.

Table 51.

Evidence Table: Gabapentinoids

Strong evidence	Evidence statement	Design
	Gabapentin is more effective than placebo in the relief of painful diabetic neuropathy and postherpetic neuralgia.	Meta analysis

Section 9.e. Antidepressants

Introduction. Antidepressants are used for the treatment of pain, dysesthesias, sleep disorders, and depression. If medications are being considered for chronic pain management, refer to the Chronic Pain Disorder MTGs medication section.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Antidepressants.

- Tricyclic antidepressants should be avoided in patients with the following:
 - at risk of unintentional overdose;
 - heart disease;
 - underlying cardiac conduction system disease;
 - dysrhythmia;
 - prostatic hypertrophy;
 - at risk for suicide;
 - uncontrolled hypertension or orthostatic hypotension;
 - those 65 years or older, particularly if a fall risk; and
 - seizure disorder.
- Serotonin norepinephrine reuptake inhibitors (SNRIs) should not be used in patients with seizures or eating disorders.

Complications of Antidepressants.

- Tricyclic antidepressants may cause acute hepatitis, neuroleptic malignant syndrome, and tardive dyskinesia.
- SNRIs may result in serotonin syndrome.
- Venlafaxine may cause hypertension, glaucoma, sexual dysfunction, and cardiac issues.

Side Effects and Adverse Events Related to Antidepressants

- Tricyclic antidepressants (e.g., amitriptyline, nortriptyline) may cause anticholinergic effects (e.g., sedation, dry mouth and associated periodontal conditions, orthostatic hypotension, constipation, urinary retention); decreased seizure threshold; sexual dysfunction; diaphoresis; tremor; antihistamine effects; alpha-1-adrenergic receptor blockade; and cardiac effects, including increasing intraventricular conduction, prolonged QT interval, prolonged conduction through the atrioventricular node.
- SNRIs (e.g., duloxetine, venlafaxine) may cause nausea, dizziness, sweating, loss of appetite, dry mouth, insomnia, drowsiness, constipation, abnormal bleeding, fatigue, sexual dysfunction, and suicidal ideation and attempts in adolescents and young adults.

Recommendations.

Core Requirements.

Recommendation 177. Tricyclic antidepressants are the recommended first-line agent for neuropathic pain, particularly in the setting of insomnia, but they are not recommended as a first-line agent for depression.

Recommendation 178. SNRIs are recommended as a second-line agent for neuropathic pain if a tricyclic offers inadequate relief. However, duloxetine may be considered a first-line agent for a patient who is a candidate for pharmacologic treatment of both chronic pain and depression.

Recommendation 179. Selective serotonin reuptake inhibitors (SSRIs) are recommended for treating depression, but they are not recommended for neuropathic pain.

Recommendation 180. Evaluation and ongoing monitoring for suicidal ideation and mood swings are required for all patients being considered for antidepressant medications.

Recommendation 181. A screening electrocardiogram may be indicated for select patients prior to initiating treatment with a tricyclic or SNRI antidepressant to assess cardiovascular risk.

Time Frames.

Time Frames for Antidepressant Medications	
Optimum duration	Maximum duration
up to 6 months	up to 12 months, with monitoring

Evidence Tables. None. The above recommendations were based on consensus.

Section 9.f. Opioids

Introduction. Opioids are powerful analgesics reserved for severe, acute pain associated with major trauma, burn, sickle cell disease, and end-of-life cancer pain. If opioid medications are being considered for chronic pain management, refer to the Chronic Pain Disorder MTGs medication section.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Opioids.

- Opioids should not be prescribed with benzodiazepines, antihistamines, or other central nervous system depressants or when the history is consistent with regular alcohol or other substance use.
- Tramadol should be used with caution in patients who have a history of seizures, are taking medications that lower the seizure threshold, or are taking medications that impact serotonin reuptake and could increase the risk for serotonin syndrome, such as monoamine oxidase inhibitors, SSRIs, selective serotonin agonists (triptans), tricyclic antidepressants, and alcohol. Also use caution in patients who are taking medications that result in QT prolongation. Tramadol has been associated with death in those with an emotional disturbance or concurrent use of alcohol and other opioids. Significant renal and hepatic dysfunction requires dosage adjustment.

Side Effects and Adverse Events Related to Opioids.

- Opioid medications may commonly cause nausea, vomiting, drowsiness, unsteadiness, constipation, and confusion. Occasional side effects include dry mouth, sweating, pruritus, hallucinations, and myoclonus. Adverse events include opioid-induced hyperalgesia, respiratory depression, dependence, opioid use disorder, overdose, or death. Prolonged opioid use may result in hypogonadism (tables 52, 53). Abrupt discontinuation may precipitate withdrawal.
- Certain comorbid medical conditions can increase the risk for opioid overdose (e.g., obesity, pulmonary disease, obstructive sleep apnea, congestive heart failure, history of alcohol or substance use disorder, advanced age, renal or hepatic dysfunction).

Recommendations.

Core Requirements.

Recommendation 182. Opioid medications are not generally recommended for the treatment of neck pain. Rare exceptions include either of the following:

- acute, severe functionally limiting pain in a patient for whom other non-opioid medications are contraindicated; or
- acute, severe functionally limiting pain that is refractory to non-opioid medications and nonpharmacologic treatment and an absence of risk factors for potential misuse or abuse ([table 54](#)).

Recommendation 183. Long-acting opioids are not recommended for the treatment of acute, subacute, or postoperative pain.

Recommendation 184. Opioid medications, including tramadol, are not generally recommended for use in patients with a history of opioid dependence. However, if an opioid medication is deemed clinically appropriate, a referral to a pain specialist is permitted.

Short-term Prescribing Requirements.

Recommendation 185. Prior to dispensing an opioid medication, the following steps are required:

- documented results of a rapid risk assessment for developing opioid use disorder (e.g., Opioid Risk Tool [ORT]),
- risk assessment of developing opioid related adverse events,
- review data on the Colorado PDMP,
- education on the short- and long-term risks and side effects of opioid therapy,
- realistic goals of opioid therapy and the anticipated course of recovery,
- establish the lowest effective dose and shortest duration of therapy,
- education on the safe storage and disposal of opioid medications, and
- develop a discontinuation plan for opioids prior to prescribing.

Prescribing Requirements Beyond 7 Days.

Recommendation 186. Whenever opioids are prescribed for more than 7 days, providers must follow all recommendations for screening and follow-up of chronic pain use. [See the Chronic Pain Disorder MTGs.](#)

Time Frames.

Time Frames for Opioids	
Optimum duration	Maximum duration
≤ 3 days	7 days [‡]
[‡] Whenever there is use of opioids for > 7 days, providers should follow all recommendations for screening and follow-ups of chronic pain use.	

Evidence Tables.

Table 52.

Evidence Table: Risk of Initiating Opioid Medications

Summary:

Short-term opioid prescriptions increase the risk of disability and the risk of developing an opioid use disorder. The use of 50 morphine milligram equivalents (MME) or greater average daily dose or the use of long-acting opioids increases the risk of opioid-related death.

Good evidence	Evidence statement	Design
	In the setting of common low back injuries, when baseline pain and injury severity are taken into account, a prescription for more than 7 days of opioids in the first 6 weeks is associated with an approximate doubling of disability 1 year after the injury.	Cohort study
	In the setting of new onset chronic noncancer pain, there is a clinically important relationship between opioid prescription and subsequent opioid use disorder. Compared to no opioid use, short-term opioid use approximately triples the risk of opioid use disorder in the next 18 months. Use of opioids for over 90 days is associated with very pronounced increased risks of the subsequent development of an opioid use disorder, which may be as much as 100-fold when doses greater than 120 MME are taken for more than 90 days. The absolute risk of these disorders is very uncertain but is likely to be greater than 6.1% for long duration treatment with a high opioid dose.	Cohort study
	In generally healthy patients with chronic musculoskeletal pain, treatment with long-acting opioids, compared to treatments with anticonvulsants or antidepressants, is associated with an increased risk of death of approximately 69%, most of which arises from non-overdose causes, principally cardiovascular in nature. The excess cardiovascular mortality principally occurs in the first 180 days from starting opioid treatment.	Cohort study
	Prescription opioids in excess of 200 MME average daily doses are associated with a near tripling of the risk of opioid-related death, compared to average daily doses of 20 MME. Average daily doses of 100-200 mg and doses of 50-99 mg per day may be associated with a doubling of mortality risk, but these risk estimates need to be replicated with larger studies.	Cohort study
Some evidence	Evidence statement	Design
	Compared to an opioid dose under 20 MME per day, a dose of 20-50 mg nearly doubles the risk of death, a dose of 50 to 100 mg may increase the risk more than fourfold, and a dose greater than 100 mg per day may increase the risk as much as sevenfold. However, the absolute risk of fatal overdose in chronic pain patients is fairly low, and may be as low as 0.04%.	Cohort study

Table 53.

Evidence Table: Adverse Effects of Opioid Medications

Strong evidence	Evidence statement	Design
	Adverse events such as constipation, dizziness, and drowsiness are more frequent with opioids than with placebo.	Systematic review
Good evidence	Evidence statement	Design
	Opioids produce significantly more adverse effects than placebo such as constipation, drowsiness, dizziness, nausea, and vomiting.	Systematic review

Table 54.

Evidence Table: Opioid Medications Effectiveness

Summary:

The majority of patients with musculoskeletal pain do not experience clinically meaningful pain relief or functional improvement with opioid treatment as compared to placebo. For patients with neuropathic pain, there is no evidence that opioid treatment improves function or quality of life. For neuropathic pain patients, opioid treatment is more likely to result in pain relief as compared to placebo, but there is no evidence that opioids are superior to gabapentin or nortriptyline for this indication. Opioids produce significantly more adverse effects than placebo. While the literature in this table discusses chronic low back pain, there is presumed physiologic similarity for neck pain.

Strong evidence	Evidence statement	Design
	In the setting of chronic nonspecific low back pain, the short and intermediate term reduction in pain intensity of opioids, compared with placebo, falls short of a clinically important level of effectiveness for a majority of patients, although some patients may experience a clinically meaningful analgesic benefit.	Systematic review
Good evidence	Evidence statement	Design
	In patients with chronic low back pain, or chronic pain from osteoarthritis of the hip and knee, opioid therapy does not lead to better pain outcomes in terms of daily functioning or in pain intensity.	RCT
	In patients with acute nontraumatic musculoskeletal low back pain who have optimized their use of NSAIDs such as naproxen, the addition of oxycodone 5 mg/acetaminophen 325 mg to be taken as needed adds no significant functional benefit in the short term, but it does increase the frequency of adverse events such as dizziness, drowsiness, and GI side effects.	RCT
	Opioids are more efficient than placebo in reducing neuropathic pain by clinically significant amounts during the first 8 weeks of treatment.	Systematic review

Lack of evidence	Lack of evidence statement	Design
	There is a lack of evidence that opioids improve function and quality of life more effectively than placebo.	Systematic review
	There is a lack of evidence that opioids are superior to gabapentin or nortriptyline for pain reduction.	Systematic review

Section 9.g. Tobacco Cessation

Introduction. Nicotine replacement therapy is used to relieve nicotine withdrawal symptoms by providing nicotine without the use of tobacco. The mechanism of bupropion in smoking cessation is not entirely understood, but it is believed to act by enhancing central nervous system noradrenergic and dopaminergic release. Varenicline reduces the symptoms of nicotine withdrawal by binding to the receptor that mediates the reinforcing effects of nicotine dependence.

Contraindications / Complications / Side Effects and Adverse Events.

Absolute and Relative Contraindications to Tobacco Cessation Therapies.

- Use nicotine replacement therapy with caution in patients with unstable cardiovascular disease.
- Bupropion is contraindicated in those with seizure disorders.

Side Effects and Adverse Events Related to Tobacco Cessation Therapies.

- Nicotine replacement therapy can cause GI symptoms, headache, and local irritation with topical products.
- Bupropion can cause insomnia, agitation, dry mouth, and headache.
- Varenicline can cause nausea, insomnia, neuropsychiatric disorders, and abnormal dreams.
- Review package insert for supplement contraindications and side effects.

Recommendations.

Core Requirements.

Recommendation 187. Tobacco cessation, including medication and behavioral support, are recommended when tobacco use is expected to negatively impact claim-related medical outcomes. Medications may include nicotine patches, gum, inhaler, lozenges or nasal spray, bupropion, or varenicline ([table 55](#)).

Evidence Table.

Table 55.		
<i>Evidence Table: Smoking and Non-Operative Spine Care</i>		
Some evidence	Evidence statement	Design
	Patients who smoke respond less favorably to non-operative spine care than nonsmokers, and quitting smoking during treatment results in greater improvement than patients who continue smoking during treatment.	Cohort study

Section 10. Interdisciplinary Rehabilitation Programs

Introduction. These guidelines discuss an interdisciplinary approach to neck pain treatment in the acute and subacute timeframes. Interdisciplinary rehabilitation programs are the gold standard of treatment for individuals with chronic neck pain who have not responded to less intensive modes of treatment or individuals who require concurrent treatment for chemical dependency.

Appendix

Functional Tests

Name of Test	Body Part	Description
Oswestry Low Back Pain Disability Questionnaire (ODI)	Lumbar Spine	Functional questions with sleeping, lifting, walking, sitting, and standing
Quebec Back Pain Disability Scale	Spine	Functional and emotional questions
Neck Disability Index	Cervical Spine	10 questions addressing pain, personal care, headaches, and functional deficits
36-Item Short Form Survey (SF-36) and 12-Item Short Form Survey (SF-12)	Physical Health	Assesses activities of daily living (ADLs)
Dallas Pain Questionnaire	Spine	Chronic pain
Brief Pain Inventory	General	Sleep, walking, ADLs

Psychological Screens

Name of Test	Description
Brief Battery for Health Improvement, 2 nd Edition (BBHI 2)	Measures pain, functioning, somatization, depression, anxiety, and defensiveness; brief measure of risk factors for delayed recovery
Distress and Risk Assessment Method (DRAM)	Measures depression and somatic symptoms of anxiety, risk factors commonly associated with chronic pain
Center for Epidemiological Studies Depression Scale (CES-D)	Measures depression, 20 items
Beck Depression Inventory-II (BDI-II)	Measures depression, 21 items
Primary Care Evaluation for Mental Disorders (PRIME-MD) <i>Must be filled out by a provider</i>	2 components: paper and pencil screen for patient and follow-up interview by physician. Assesses mood, anxiety, somatoform tendencies, and alcohol and eating disorders
Zung Depression Inventory	Measures depression, brief measure
Patient Health Questionnaire (PHQ) and PHQ-9	Self-administered version of the PRIME-MD. Assesses mood, anxiety, somatoform tendencies, and alcohol and eating disorders
Generalized Anxiety Disorder Scale (GAD-7)	Assesses generalized anxiety, 7 questions
Behavioral Health Index-Multimedia Version (BHI-MV)	Screens for addiction

Editor's Notes

7 CCR 1101-3 has been divided into smaller sections for ease of use. Versions prior to 01/01/2011, and rule history, are located in the first section, 7 CCR 1101-3. Prior versions can be accessed from the History link that appears above the text in 7 CCR 1101-3. To view versions effective after 01/01/2011, select the desired part of the rule, for example 7 CCR 1101-3 Rules 1-17, or 7 CCR 1101-3 Rule 18: Exhibit 1.

History

[For history of this section, see Editor's Notes in the first section, 7 CCR 1101-3]